



Lecture
Notes

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Approximation Methods

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1 Introduction: Order-of-Magnitude Estimation

1.1 Motivation

In most physics courses, the problem is already pre-packaged for you: the relevant physics is identified, the boundary conditions are given, and the main challenge is to carry out the correct mathematical procedure. Real life is completely different.

In practice—whether you are designing an experiment for your PhD, evaluating whether cracks in a hospital building are dangerous, or checking if the vibrations of an approaching light-rail will disturb a precision magnetic measurement—you are given a raw situation and asked: *does this matter?* You need to identify the relevant physics, estimate which effects dominate, and obtain a reliable answer without solving the full problem from first principles.

This is the art of **order-of-magnitude (Fermi) estimation**: a set of reasoning principles that let a physicist quickly obtain a result accurate to within one or two orders of magnitude, which is usually enough to decide whether something is important.

Remark 1 (Enrico Fermi). Fermi was famous for such problems. He would ask his students, for example: “*How many piano tuners are there in Chicago?*” The point is not whimsy—it is that a physicist can estimate this reliably by decomposing the problem: city population $\rightarrow$ fraction with pianos $\rightarrow$ tuning frequency $\rightarrow$ time per tuning $\rightarrow$ number of tuners. The same decomposition principle underlies every example in this lecture.

1.2 Principle 1: Divide and Conquer

The first and most important strategy is to **decompose** a seemingly intractable problem into a product (or ratio) of simpler sub-quantities, each of which can be estimated independently.

$$\text{Answer} = Q_1 \times Q_2 \times Q_3 \times \cdots$$

Each Q_i may carry its own uncertainty, but because the factors multiply, dimensional analysis and unit tracking help catch algebraic errors. After estimating each factor, collect the results and read off the order of magnitude.

Worked Example 1.1 (Diaper Landfills — Hiriya)

Question. How many landfills the size of the Hiriya (Park Ariel Sharon, near Tel Aviv) are filled each year by the diapers of American babies?

Decomposition.

We identify four building blocks:

1. N_{baby} : number of American babies
2. D_{yr} : diapers per baby per year
3. V_{diaper} : volume of one (used) diaper
4. V_{H} : volume of the Hiriya landfill

Step 1: Number of American babies.

The US population is approximately $N_{\text{total}} \approx 4 \times 10^8$ people. For a developed country with a life expectancy of ~ 80 years and a roughly uniform age distribution, the fraction who are under 2 years old (the diaper-wearing period) is:

$$f_{\text{baby}} = \frac{2 \text{ yr}}{80 \text{ yr}} = \frac{1}{40}.$$

Therefore:

$$N_{\text{baby}} = N_{\text{total}} \times f_{\text{baby}} \approx 4 \times 10^8 \times \frac{1}{40} = 10^7 \text{ babies.}$$

Step 2: Diapers per baby per year.

A newborn uses roughly 4 diapers per day:

$$D_{\text{yr}} \approx 4 \frac{\text{diapers}}{\text{day}} \times 300 \text{ days/yr} \approx 10^3 \frac{\text{diapers}}{\text{baby} \cdot \text{yr}}.$$

(Using 300 instead of 365 for arithmetic convenience; the error is $\lesssim 20\%$, well within our target precision.)

Step 3: Volume of one diaper.

Empirically (from a standard supermarket pack): a package of 50 diapers measures roughly $50 \text{ cm} \times 20 \text{ cm} \times 10 \text{ cm}$.

$$V_{\text{pack}} = 50 \times 20 \times 10 = 10,000 \text{ cm}^3.$$

Volume per diaper (compressed, as sold):

$$V_{\text{dry}} = \frac{10,000}{50} = 200 \text{ cm}^3 \text{ per diaper.}$$

When soiled and deposited in a landfill, the diaper expands slightly but most moisture evaporates; we estimate the effective deposited volume at:

$$V_{\text{diaper}} \approx 300 \text{ cm}^3.$$

Step 4: Volume of the Hiriya.

The Hiriya is a roughly conical hill with base diameter $\sim 600 \text{ m}$ and height $\sim 60 \text{ m}$ (the lecturer and students debated the height; Google Earth values are around 60 m). Treating it as a cone:

$$V_H = \frac{1}{3} \pi r^2 h \approx \frac{1}{3} \times 300^2 \times 60 \approx \frac{\pi}{3} (300)^2 (60).$$

For a quick estimate we drop the $\pi/3 \approx 1$ factor:

$$V_H \approx (600 \text{ m})^2 \times 60 \text{ m} \times \frac{1}{3} \approx 600^2 \times 60/3 \text{ m}^3.$$

Numerically: $600^2 = 3.6 \times 10^5$, so $V_H \approx 3.6 \times 10^5 \times 20 \approx 7 \times 10^6 \text{ m}^3$. To keep a round number we use:

$$V_H \approx 2 \times 10^7 \text{ m}^3.$$

Assembling the result.

The total diaper volume generated in the US per year is:

$$\dot{V}_{\text{diapers}} = N_{\text{baby}} \times D_{\text{yr}} \times V_{\text{diaper}} = 10^7 \text{ baby} \times 10^3 \frac{\text{diapers}}{\text{baby} \cdot \text{yr}} \times 300 \frac{\text{cm}^3}{\text{diaper}}.$$

Units cancel perfectly: baby $\times$ diaper / baby = diaper, diaper $\times$ cm³ / diaper = cm³.

$$\dot{V}_{\text{diapers}} = 3 \times 10^{12} \frac{\text{cm}^3}{\text{yr}}.$$

Converting to m³ (1 m = 100 cm, so 1 m³ = 10⁶ cm³):

$$\dot{V}_{\text{diapers}} = \frac{3 \times 10^{12}}{10^6} \frac{\text{m}^3}{\text{yr}} = 3 \times 10^6 \frac{\text{m}^3}{\text{yr}}.$$

The number of Hiriya-equivalents filled per year:

$$n_H = \frac{\dot{V}_{\text{diapers}}}{V_H} = \frac{3 \times 10^6}{2 \times 10^7} \text{ yr}^{-1} \approx 0.15 \text{ yr}^{-1},$$

i.e. it takes approximately $1/0.15 \approx 6$ years for American diapers to fill one Hiriya.

1.3 Principle 2: Cross-Check via an Independent Route

A single estimate is useful; *two independent estimates of the same quantity is much more powerful*. If both routes agree within an order of magnitude, you have good confidence in the answer. If they disagree by more than two orders of magnitude, you have almost certainly made a conceptual error (wrong model, wrong assumption) somewhere—and finding it teaches you something deep about the problem.

Worked Example 1.2 (Hiriya cross-check via human-waste accounting)

Instead of tracking diaper volume directly, we can estimate the Hiriya's volume in units of *person-years of garbage*.

Hiriya in person-years:

The Hiriya accumulated garbage from the Gush Dan region (greater Tel Aviv area) for ~ 50 years. The population grew from $\sim 200,000$ to $\sim 1,000,000$ over that period; a reasonable average is $\bar{P} \approx 5 \times 10^5$ people:

$$\text{person-years in Hiriya} = \bar{P} \times t = 5 \times 10^5 \times 50 = 2.5 \times 10^7 \text{ person-yr.}$$

Diaper fraction of garbage:

A family of 4 produces ~ 1 garbage bag per day. If the family has a baby using 4 diapers per day, each diaper has roughly the same volume as $\frac{1}{4}$ of a garbage bag. So diapers are $\sim \frac{4}{4 \times (\text{bags per bag})}$ of the family's garbage. More carefully: 4 diapers $\approx \frac{1}{2}$ bag, total garbage ≈ 1 bag $\Rightarrow$ diaper fraction $\approx \frac{1}{2}$. But only the fraction of the population that has a baby uses diapers. For a family of 4 with a baby present for

2 of 80 years:

$$f_{\text{diap}} \approx \frac{1}{2} (\text{diaper fraction while baby present}) \times \frac{2}{80} (\text{fraction of family lifetime}) = \frac{1}{80} \approx 0.012.$$

Alternatively: one baby-year corresponds to one out of the family's $80/4 = 20$ person-years where a baby is present, and diapers are $\sim \frac{1}{2}$ of the waste during that time:

$$f_{\text{diap}} \approx \frac{1}{2} \times \frac{1}{20} = \frac{1}{40} \approx 0.025.$$

We take $f_{\text{diap}} \approx \frac{1}{20}$.

US population equivalent:

$N_{\text{US}} = 4 \times 10^8$ persons. In the US, diapers accumulate at a rate of:

$$\dot{V}_{\text{diapers}}^{(2)} = \frac{\text{person-year of Hiriya}}{N_{\text{US}}} \times f_{\text{diap}}^{-1} \times V_H.$$

More directly: the time to fill Hiriya is:

$$\tau = \frac{\text{person-yr of Hiriya} \times f_{\text{diap}}}{N_{\text{US}}} = \frac{2.5 \times 10^7 \times \frac{1}{20}}{4 \times 10^8} = \frac{1.25 \times 10^6}{4 \times 10^8} \approx 3 \times 10^{-3} \text{ yr.}$$

Remark 2. This gives ~ 1 day—clearly too short! The lecturer used this discrepancy to illustrate that the diaper fraction f_{diap} here acts in the *wrong direction*: a smaller f_{diap} means diapers constitute a smaller fraction of waste, so it takes *longer*, not shorter, to fill Hiriya with diapers. Correcting the inversion:

$$\tau_{\text{corrected}} = \frac{2.5 \times 10^7}{4 \times 10^8 \times f_{\text{diap}}} = \frac{2.5 \times 10^7}{4 \times 10^8 \times \frac{1}{20}} = \frac{2.5 \times 10^7}{2 \times 10^7} \approx 1 \text{ yr.}$$

This crosses route 1's answer of ~ 6 years within a factor of 6, which is excellent for a Fermi estimate. The lesson: **always track sign/direction carefully**, and the fact that the two routes agreed within a factor of 6 (not 6 orders of magnitude) means the model is self-consistent.

Worked Example 1.3 (Meteor Impact Rate on Earth)

Question. How many meteorites hit the Earth per year?

Key data. Throughout all of recorded human history there are exactly **two** documented cases of a meteorite striking a living creature: one woman (bruised, Alabama, USA, 1954) and one dog (killed, Egypt, 1911). We can use this fact together with a model of human population growth to infer the meteorite flux.

Strategy.

We compute the effective *total cross-section-time* exposed by humanity, $\Sigma_{\text{HR}} [\text{m}^2 \cdot \text{yr}]$, and then extract the meteorite flux $\Phi [\text{m}^{-2} \text{yr}^{-1}]$ from:

$$\Phi \times \Sigma_{\text{HR}} \approx 1 \quad (\text{roughly 1 recorded strike on a human}).$$

Model for human population growth.

The human population has grown roughly exponentially. Let $N(t)$ be the population at time t before the present (t measured in years, increasing into the past). We write:

$$N(t) = N_0 e^{-t/\tau},$$

where $N_0 \approx 8 \times 10^9$ is today's population and $\tau \approx 70$ yr is the population doubling-time-related scale (the population roughly doubled in the last 70 years, consistent with $\tau \sim 70/\ln 2 \approx 100$ yr; we use $\tau = 70$ yr as a round number).

The total person-years of history is dominated by recent centuries:

$$\text{PY} = \int_0^\infty N_0 e^{-t/\tau} dt = N_0 \tau \approx 8 \times 10^9 \times 70 \approx 5.6 \times 10^{11} \text{ person-yr.}$$

Human cross-section and total exposure.

The cross-sectional area of one person (projected onto the ground) is:

$$\sigma_1 \approx \frac{1}{2} \text{ m}^2.$$

Therefore the total human cross-section-time is:

$$\Sigma_{\text{HR}} = \sigma_1 \times \text{PY} \approx \frac{1}{2} \times 5.6 \times 10^{11} \approx 2.8 \times 10^{11} \text{ m}^2 \cdot \text{yr.}$$

Inferred meteorite flux.

$$\Phi = \frac{1 \text{ event}}{\Sigma_{\text{HR}}} \approx \frac{1}{2.8 \times 10^{11}} \approx \frac{1}{3 \times 10^{11}} \text{ m}^{-2} \text{ yr}^{-1}.$$

Impacts on all of Earth.

The surface area of the Earth is:

$$A_\oplus = 4\pi R_\oplus^2, \quad R_\oplus \approx 6.4 \times 10^6 \text{ m.}$$

$$A_\oplus \approx 4 \times 3 \times (6.4)^2 \times 10^{12} \approx 5.1 \times 10^{14} \text{ m}^2.$$

Annual number of meteorite impacts on Earth:

$$\dot{N}_{\text{impacts}} = \Phi \times A_\oplus \approx \frac{5.1 \times 10^{14}}{3 \times 10^{11}} \approx \frac{5 \times 10^{14}}{3 \times 10^{11}} \approx \boxed{1,500 \text{ meteorites per year.}}$$

Remark 3. Humans cover only a tiny fraction of Earth's surface ($\sim 10^{-4}$), so the flux inferred from the 1 human-strike event maps onto $\sim 1,500$ total Earth impacts per year. This result is consistent with observational estimates. The key physical insights are:

1. Population growth is exponential, so the *effective* person-year baseline is just $N_0 \tau$ —almost all of humanity's exposure has happened in modern times.
2. A simple exponential model for a complex historical process is nonetheless a good starting point.

Worked Example 1.4 (Is it Worth Robbing a Brink's Truck?)

Question. How much cash does a Brink's armored vehicle typically carry? Is there enough to make a heist financially worthwhile?

Method 1: Weight approach.

Step 1: Value-to-weight ratio of cash.

Take 40 bills and fold them twice: the stack fits in $\sim 0.5 \text{ cm} \times 5 \text{ cm} \times 10 \text{ cm}$:

$$V_{40 \text{ bills}} \approx 0.5 \times 5 \times 10 = 25 \text{ cm}^3.$$

A single bill therefore occupies:

$$v_{\text{bill}} \approx \frac{25}{40} \approx 0.6 \text{ cm}^3.$$

Paper has density $\rho \approx 1 \text{ g/cm}^3$, so one bill weighs $\sim 0.6 \text{ g}$. A typical banknote in Israel is 200 NIS $\approx 50 \text{ USD}$. The value-to-mass ratio:

$$\frac{\text{value}}{\text{mass}} \approx \frac{200 \text{ NIS}}{0.6 \text{ g}} \approx 330 \frac{\text{NIS}}{\text{g}} \approx 80 \frac{\text{USD}}{\text{g}}.$$

Remark 4 (Curious equivalence with gold). Gold's current price is $\sim \$3,000$ per troy ounce ($\approx 30 \text{ g}$), giving:

$$\frac{\text{gold value}}{\text{mass}} \approx \frac{3000}{30} = 100 \frac{\text{USD}}{\text{g}}.$$

Remarkably, **cash and gold have nearly the same value per unit mass**—which means it is no more efficient to carry gold than banknotes in a heist!

Step 2: Payload of the truck.

One duffel bag of cash weighs $\sim 10 \text{ kg}$. A typical Brink's run might involve a few bags:

$$\text{Cash} \approx \text{few bags} \times 10 \text{ kg/bag} \times \frac{80 \text{ USD}}{\text{g}} = \text{few} \times 10 \times 10^3 \times 80 \text{ USD} = \text{few} \times 800\,000 \text{ USD}.$$

So one bag already represents $\sim \$800,000$.

Method 2: Bank-turnover approach (Equipartition argument).

A branch bank serves a neighbourhood. Consider a city of 25,000, with 3 bank branches, so each branch has $\sim 8,000$ customers ($\sim 2,500$ households).

Weekly cash withdrawal per household: roughly 1000 NIS $\approx \$250$ (for expenses paid in cash: groceries market, domestic help, etc.).

Total cash distributed by one branch per week:

$$C_{\text{out}} \approx 2,500 \text{ households} \times \frac{\$250}{\text{household} \cdot \text{week}} = \$625,000 \text{ per week}.$$

Equipartition principle: The cash that private customers withdraw must eventually return through business deposits (shop owners, market vendors, service providers). In steady state (no net accumulation under the mattresses!) the cash-in rate equals

the cash-out rate. Therefore the Brink's truck, which services the branch roughly once per week, carries $\sim \$600,000$ — consistent with Method 1.

Conclusion.

Both methods point to **several hundred thousand dollars per Brink's run**. Whether that is “worth it” depends on the probability of being caught—a calculation left to the student. The deeper lesson for physics is the **equipartition / steady-state argument**: whenever two coupled flows must balance in equilibrium, each flow carries roughly the same quantity.

1.4 Principle 3: Equipartition

The Brink's example illustrates a general physics principle that reappears throughout the course:

Physical Concept 1.1: Equipartition of coupled quantities

When two (or more) physical processes are coupled by exchange, and the system reaches steady state (or thermodynamic equilibrium), the relevant energy or quantity is *shared roughly equally* among the coupled degrees of freedom.

Examples from physics:

- **Virial theorem:** For a gravitationally or Coulomb-bound system, $\langle T \rangle = -\frac{1}{2}\langle U \rangle$, so $\langle T \rangle$ and $|\langle U \rangle|$ are of the same order.
- **Interstellar medium (ISM):** The ISM contains magnetic fields, thermal gas, turbulent kinetic energy, and cosmic rays. Because each couples to the others (shock waves accelerate cosmic rays; cosmic rays exert pressure on magnetic fields; turbulence amplifies magnetic fields), all four energy densities are within an order of magnitude of each other: $\sim 1 \text{ eV cm}^{-3} \sim 10^{-12} \text{ erg cm}^{-3}$.
- **Cash in a bank:** Private withdrawals $\approx$ business deposits in steady state.

1.5 Preview: The Shooting Star (to be continued)

At the end of the lecture, the following problem was introduced and will be completed in the next session:

Question. When you see a shooting star (meteor), what is its mass?

The strategy: compare the brightness of the meteor to a known source (e.g., a street-lamp at 10 km). The meteor travels at $v \sim 30 \text{ km/s}$ (of order the Earth's orbital velocity around the Sun); the kinetic energy $\frac{1}{2}mv^2$ is partially converted into light. Estimating what fraction becomes visible light, and matching to the observed brightness, will yield the mass.

Remark 5. Speeds of order 30 km/s are natural because a typical meteoroid starts at rest relative to the Sun at some distance, falls toward Earth, and arrives approximately at the escape velocity from the Sun at Earth's orbit—which turns out to be $\sim \sqrt{2}v_{\oplus} \approx 42 \text{ km/s}$. Orbiting meteors reach Earth's frame at $|v_{\oplus} \pm v_{\text{meteor}}|$, so $\sim 10\text{--}70 \text{ km/s}$ is the expected range.

Summary of Principles Introduced in Lecture 1

1. **Divide and conquer:** decompose the problem into independently estimable factors and multiply them together. Track units to catch errors.
2. **Cross-check with an independent route:** if two very different derivations agree, the result is probably right; if they disagree by $\gg 10^2$, find the conceptual error.
3. **Equipartition / steady-state:** coupled processes tend to equilibrate, so the shared quantities are of the same order of magnitude.

2 Lecture 2: Scaling and Approximation

In this lecture, we conclude the examples of order-of-magnitude estimation (specifically, the shooting star problem) and introduce a new, extremely powerful tool: **Scaling Arguments**. By comparing a physical system either to another known system or to a different state of itself, we can cancel out many unknown factors and gain deep physical insights.

2.1 Completing the Shooting Star Estimate

Continuing from the previous lecture, we aim to estimate the mass of a typical “shooting star” (meteor).

Worked Example 2.1 (Mass of a Shooting Star)

The Setup: When a meteoroid enters Earth’s atmosphere, it burns up, converting its kinetic energy into heat and light. We want to estimate its mass M by observing the visible light it emits.

Observable Properties:

1. **Duration:** The event happens very quickly, typically taking $\Delta t \sim 0.5\text{--}1$ second.
2. **Velocity:** As discussed previously, the typical entry velocity is on the scale of Earth’s orbital speed around the Sun: $V \sim 30\text{ km/s} = 3 \times 10^4\text{ m/s}$.
3. **Brightness (Luminosity):** To estimate the luminous power, we can compare it to a familiar object, such as a car headlight seen from a distance. Suppose a shooting star appears as bright as a car’s headlight ($P_{\text{ref}} \sim 10\text{ W}$ of *pure optical power*) viewed from a distance of $D_{\text{ref}} \sim 3\text{ km}$.
4. **Distance to Meteor:** The meteor typically burns up in the upper atmosphere, at an altitude $L \sim 50\text{ km}$ to 100 km . For our estimate, let us use $L \sim 100\text{ km} = 10^5\text{ m}$.

Energy Conversion Model: The initial kinetic energy is $E_k = \frac{1}{2}MV^2$. By the principle of equipartition (assuming roughly equal partition of the dissipated energy into thermal energy of the plasma and radiated light), the energy radiated as light is a fraction $f_{\text{light}} \sim \frac{1}{2}$ of the total energy. Thus, the total light energy emitted is:

$$E_{\text{light}} = f_{\text{light}} \left(\frac{1}{2}MV^2 \right) \approx \frac{1}{4}MV^2$$

The average optical power emitted during the time Δt is:

$$P_{\text{meteor}} = \frac{E_{\text{light}}}{\Delta t} = \frac{MV^2}{4\Delta t}$$

Flux Comparison: We observe the flux (power per unit area). Assuming the light is emitted isotropically, the flux we observe from the meteor at distance L is matched to the flux of the reference headlight at distance D_{ref} :

$$F_{\text{obs}} = \frac{P_{\text{meteor}}}{4\pi L^2} \approx \frac{P_{\text{ref}}}{4\pi D_{\text{ref}}^2} = F_{\text{ref}}$$

Substituting P_{meteor} :

$$\frac{MV^2}{4\Delta t \cdot 4\pi L^2} \approx \frac{P_{\text{ref}}}{4\pi D_{\text{ref}}^2} \implies \frac{MV^2}{4\Delta t \cdot L^2} \approx \frac{P_{\text{ref}}}{D_{\text{ref}}^2}$$

Solving for M :

$$M \approx \frac{4\Delta t \cdot P_{\text{ref}}}{V^2} \left(\frac{L}{D_{\text{ref}}} \right)^2$$

Calculation: Plugging in our rough numbers ($\Delta t = 0.5$ s, $P_{\text{ref}} = 10$ W, $V = 3 \times 10^4$ m/s, $L = 100$ km, $D_{\text{ref}} = 3$ km):

$$\left(\frac{L}{D_{\text{ref}}} \right)^2 \approx \left(\frac{100}{3} \right)^2 \approx 1000$$

$$M \approx \frac{4 \times 0.5 \times 10}{(3 \times 10^4)^2} \times 1000 = \frac{20}{9 \times 10^8} \times 1000 \approx 2 \times 10^{-5} \text{ kg} = 20 \text{ mg}.$$

Remark 6. The result is startling: the brilliant streak across the night sky is caused not by a massive rock, but by a tiny grain of dust weighing barely a few milligrams. The enormous optical power derives entirely from the extremely high velocity ($V^2 \propto 10^9$)!

2.2 Principle 4: Scaling Arguments

When making approximations, a single direct calculation often propagates huge uncertainties (e.g., getting a duration wrong by a factor of 2 leads to a significant error). **Scaling** allows us to bypass this by computing the *ratio* of related quantities. This relies on referencing an unknown system against a known one, gracefully canceling out parameters we do not know.

Worked Example 2.2 (Estimating Gravity on the Moon)

Question: Suppose we want to estimate the acceleration of gravity on the Moon, g_{moon} , relative to Earth's $g_{\oplus} \approx 10$ m/s², without looking up its mass or radius.

From Newton's Law of Universal Gravitation, $g = \frac{GM}{R^2}$. Taking the ratio against Earth:

$$\frac{g_{\text{moon}}}{g_{\oplus}} = \frac{GM_{\text{moon}}/R_{\text{moon}}^2}{GM_{\oplus}/R_{\oplus}^2} = \frac{M_{\text{moon}}}{M_{\oplus}} \left(\frac{R_{\oplus}}{R_{\text{moon}}} \right)^2$$

Since mass $M \propto \rho R^3$, we can rewrite this ratio in terms of density (ρ):

$$\frac{g_{\text{moon}}}{g_{\oplus}} = \frac{\rho_{\text{moon}}}{\rho_{\oplus}} \left(\frac{R_{\text{moon}}}{R_{\oplus}} \right)^3 \left(\frac{R_{\oplus}}{R_{\text{moon}}} \right)^2 = \frac{\rho_{\text{moon}}}{\rho_{\oplus}} \frac{R_{\text{moon}}}{R_{\oplus}}$$

Notice that the gravitational constant G has completely canceled out. We now just need the density ratio and the radius ratio.

Estimating the Radius Ratio: We can measure the angular size of the Moon, θ_{moon} . A finger at arm's length subtends roughly 1.5° , or $\approx 1/40$ radians. The Moon is about one-third of a finger width, yielding:

$$\theta_{\text{moon}} \approx \frac{1}{120} \text{ rad.}$$

The radius of the Moon is $R_{\text{moon}} \approx \frac{1}{2} \theta_{\text{moon}} D_{\text{moon}}$, where D_{moon} is the distance to the Moon. How do we find D_{moon} in terms of Earth's radius $R_{\oplus}$? We use Kepler's Third Law ($T \propto R^{3/2}$) and compare the Moon to a Low Earth Orbit (LEO) satellite:

$$\frac{T_{\text{moon}}}{T_{\text{LEO}}} = \left(\frac{D_{\text{moon}}}{R_{\text{LEO}}} \right)^{3/2}$$

An LEO satellite orbits just above Earth's surface ($R_{\text{LEO}} \approx R_{\oplus}$) with a period $T_{\text{LEO}} \approx 90$ min. The Moon's orbital period is ≈ 30 days.

$$\left(\frac{D_{\text{moon}}}{R_{\oplus}} \right)^{3/2} \approx \frac{30 \times 24 \times 60 \text{ min}}{90 \text{ min}} \approx 480$$

$$\frac{D_{\text{moon}}}{R_{\oplus}} \approx 480^{2/3} \approx 60.$$

Meaning the distance to the Moon is roughly 60 Earth radii. Now we return to the radius of the Moon:

$$R_{\text{moon}} = \frac{1}{2} \theta_{\text{moon}} D_{\text{moon}} = \frac{1}{2} \left(\frac{1}{120} \right) (60 R_{\oplus}) = \frac{1}{4} R_{\oplus}.$$

So the radius of the Moon is about $1/4$ that of Earth (the true value is $\approx 1/3.7$).

Determining the Density Ratio: If we tentatively assume the Moon has the same density as Earth ($\rho_{\text{moon}} \approx \rho_{\oplus}$), our formula gives:

$$\frac{g_{\text{moon}}}{g_{\oplus}} \approx 1 \times \frac{1}{4} = 0.25$$

However, we know experimentally that $g_{\text{moon}} \approx \frac{1}{6} g_{\oplus} \approx 0.16$. The discrepancy signals a physical reality: **The Moon is less dense than Earth**. This happens because Earth has a dense iron core, whereas the Moon is primarily composed of lighter mantle-like material (supporting the giant-impact hypothesis for the Moon's formation). By scaling, we deduced an internal geophysical property purely from simple kinematic observations!

Worked Example 2.3 (Tidal Forces: Sun vs Moon)

Question: Which exerts a stronger tidal force on Earth's oceans: the Sun or the Moon?

The tidal force (or more precisely, the differential gravitational acceleration) on the surface of Earth due to a body of mass M at distance D is given by the difference in gravity across Earth's radius $R_\oplus$:

$$a_{\text{tidal}} \approx \frac{GM}{(D - R_\oplus)^2} - \frac{GM}{D^2} \approx \frac{2GMR_\oplus}{D^3}.$$

Let us take the ratio of the Sun's tidal force to the Moon's tidal force:

$$\frac{a_{\text{tidal, Sun}}}{a_{\text{tidal, Moon}}} = \frac{2GM_{\text{sun}}R_\oplus/D_{\text{sun}}^3}{2GM_{\text{moon}}R_\oplus/D_{\text{moon}}^3} = \frac{M_{\text{sun}}}{M_{\text{moon}}} \left(\frac{D_{\text{moon}}}{D_{\text{sun}}} \right)^3.$$

Here we use an elegant cosmic coincidence: **the angular sizes of the Sun and the Moon in the sky are nearly identical** (which is why total solar eclipses exist).

$$\theta_{\text{sun}} \approx \theta_{\text{moon}} \implies \frac{R_{\text{sun}}}{D_{\text{sun}}} \approx \frac{R_{\text{moon}}}{D_{\text{moon}}} \implies \left(\frac{D_{\text{moon}}}{D_{\text{sun}}} \right)^3 \approx \left(\frac{R_{\text{moon}}}{R_{\text{sun}}} \right)^3$$

Substituting this back into the tidal ratio equation:

$$\frac{a_{\text{tidal, Sun}}}{a_{\text{tidal, Moon}}} \approx \frac{M_{\text{sun}}}{M_{\text{moon}}} \left(\frac{R_{\text{moon}}}{R_{\text{sun}}} \right)^3 = \frac{M_{\text{sun}}/R_{\text{sun}}^3}{M_{\text{moon}}/R_{\text{moon}}^3} = \frac{\rho_{\text{sun}}}{\rho_{\text{moon}}}.$$

This highlights the immense power of scaling arguments: **The ratio of their tidal forces depends solely on the ratio of their average densities!**

We know empirically from observing spring and neap tides that the solar tide is roughly half the amplitude of the lunar tide ($a_{\text{tidal, Sun}} \approx 0.5 \cdot a_{\text{tidal, Moon}}$). Consequently, we can conclude that the Sun's average density is about half of the Moon's density:

$$\rho_{\text{sun}} \approx \frac{1}{2} \rho_{\text{moon}}.$$

Since the Moon's density is mostly rock ($\rho_{\text{moon}} \approx 3 \text{ g/cm}^3$), the average density of the Sun must be $\approx 1.5 \text{ g/cm}^3$ (just slightly denser than liquid water).

2.3 Preview: Lunar Retreat and Earth's Spin

Looking ahead, we can analyze the macroscopic dynamics of the Earth-Moon system. Because Earth rotates faster than the Moon orbits (1 day vs 30 days), the tidal bulges raised by the Moon are dragged slightly ahead of the Earth-Moon axis. The gravitational pull of the Moon on this misaligned bulge creates a torque that slows Earth's rotation while simultaneously transferring angular momentum to the Moon, causing its orbit to expand.

Current laser ranging measurements (using retroreflectors left by Apollo missions)

indicate the Moon is receding at $v \approx 3.8 \text{ cm/yr}$. If we crudely extrapolate this linearly:

$$\text{Time to reach current distance } D_{\text{moon}} \approx \frac{380\,000 \text{ km}}{3.8 \text{ cm/yr}} = \frac{3.8 \times 10^{10} \text{ cm}}{3.8 \text{ cm/yr}} \approx 10^{10} \text{ years.}$$

This naive linear estimate yields 10 billion years. However, Earth is only about 4.5 billion years old! The resolution to this paradox is that the tidal interaction is highly non-linear with distance; when the Moon was closer, the tidal forces (which scale as $1/D^3$) were vastly stronger, causing a much faster recession rate. We will formulate the differential equation for this retreat in the next lecture.

3 Lecture 3: Advanced Scaling and Dimensional Analysis

3.1 Scaling of the Earth-Moon System: Lunar Retreat

In the previous lecture, we established that the Moon's density is lower than Earth's by examining tidal forces. We now turn to the *dynamics* of the Earth-Moon system.

The Physical Picture: Earth rotates on its axis once every 24 hours, while the Moon orbits Earth roughly once every 30 days. Because Earth's rotation is faster than the Moon's orbit, the tidal bulges raised on Earth's oceans are dragged ahead of the Earth-Moon axis. The Moon exerts a gravitational pull on this misaligned bulge, which produces a torque. This torque:

- Slows down Earth's rotation (lengthening the day).
- Transfers angular momentum to the Moon, causing it to retreat.

Laser ranging measurements reveal that the Moon is currently receding at a velocity of $v_0 \approx 3.8 \text{ cm/yr}$.

Mathematical Modeling via Scaling: We wish to model the distance to the Moon, $r(t)$, without solving the full Navier-Stokes equations for the oceans. We rely on angular momentum and scaling.

The orbital angular momentum of the Moon is $L = M_{\text{moon}} v_{\text{orbit}} r$. Using Kepler's laws, $v_{\text{orbit}} = \sqrt{GM_{\oplus}/r}$, so:

$$L \propto r^{1/2}$$

The torque τ exerted by the tidal bulge is equal to the rate of change of angular momentum: $\tau = dL/dt$. How does this torque scale with r ? The tidal force scales as $F_{\text{tide}} \propto 1/r^3$. The height of the tidal bulge also scales as $1/r^3$ (in a linear regime), meaning the mass of the water in the bulge scales as $1/r^3$. The torque is proportional to the product of the tidal force and the mass of the bulge, yielding an extreme sensitivity to distance:

$$\tau \propto \frac{1}{r^6} \implies \frac{dL}{dt} \propto r^{-6}$$

Let us generalize this by writing $\frac{dL}{dt} \propto r^{-q}$, where q is expected to be around 6, but must be $q > 3$.

Substituting $L \propto r^{1/2}$:

$$\frac{d}{dt} (r^{1/2}) \propto r^{-q} \implies \frac{1}{2} r^{-1/2} \frac{dr}{dt} \propto r^{-q} \implies \frac{dr}{dt} \propto r^{-(q-1/2)}$$

We can separate variables and integrate:

$$r^{q-1/2} dr \propto dt \implies \int_{r_{\text{initial}}}^{r_{\text{today}}} r^{q-1/2} dr \propto \int_0^T dt$$

Assuming the initial distance was negligible compared to today's distance ($r_{\text{initial}} \ll r_{\text{today}}$), we obtain:

$$r_{\text{today}}^{q+1/2} \propto T \implies T \sim \frac{r_{\text{today}}}{(q+1/2) \left(\frac{dr}{dt}\right)_{\text{today}}}$$

where $\left(\frac{dr}{dt}\right)_{\text{today}} = v_0 = 3.8 \text{ cm/yr}$.

The Lunar Paradox: A naive linear extrapolation ($v_0 = \text{const}$) gives an age of:

$$T_{\text{linear}} = \frac{r_{\text{today}}}{v_0} \approx \frac{3.8 \times 10^{10} \text{ cm}}{3.8 \text{ cm/yr}} \approx 10^{10} \text{ years.}$$

However, using our non-linear scaling with $q = 6$, the actual time since the Moon was close to Earth is:

$$T = \frac{T_{\text{linear}}}{6+1/2} = \frac{10^{10}}{6.5} \approx 1.5 \times 10^9 \text{ years.}$$

This presents a paradox! Earth is 4.5 billion years old, yet this model suggests the Moon formed only 1.5 billion years ago. The resolution lies in the fact that tidal friction is currently *anomalously high*. Tidal dissipation primarily occurs in shallow continental shelves, and the present configuration of Earth's continents provides more shallow seas than the historical average.

3.2 Scaling in Biology: Why All Animals Jump to the Same Height

A striking observation in biology is that animals of vastly different sizes—from a mouse to a human to a horse—can all jump to a roughly similar height of $h \sim 1 \text{ m}$. Why?

The Physics of Jumping: Jumping is an anaerobic process powered by the rapid release of energy stored in muscles (ATP). Let us build a simple model:

1. **Available Energy:** The stored energy E_{stored} is proportional to the mass of the jumping muscles. If M is the total mass of the animal, the muscle mass is $\epsilon_m M$, where ϵ_m is the fraction of body mass dedicated to jumping. Assuming the energy density ρ_E (energy per unit mass) is roughly constant across mammalian biology, the total available energy is:

$$E_{\text{stored}} = \rho_E \epsilon_m M \propto M$$

2. **Required Energy:** To elevate its center of mass by a height h , the animal must expend potential energy:

$$E_p = Mgh$$

Equating the two yields:

$$Mgh \approx \rho_E \epsilon_m M \implies h \approx \frac{\rho_E \epsilon_m}{g}$$

The total mass M *cancels out*! Because all mammals share similar muscle biology (ρ_E and ϵ_m), the maximum jump height h is independent of the animal's size.

When Does the Model Break Down? A flea (~ 1 mm) cannot jump 1 m. At very small scales, air resistance (drag) becomes a dominant factor. The work done against drag is $W_{\text{drag}} = F_{\text{drag}} \cdot h$. The drag force is $F_{\text{drag}} \sim \rho_{\text{air}} v^2 A$, where A is the cross-sectional area and v is the initial velocity. Since $v \sim \sqrt{gh}$, we have $F_{\text{drag}} \sim \rho_{\text{air}} ghA$. The cross-sectional area A relates to the animal's dimension R as $A \sim R^2$. Since $M \sim \rho_{\text{water}} R^3$, we can express the ratio of energy lost to drag vs. potential energy:

$$\frac{W_{\text{drag}}}{E_p} \sim \frac{\rho_{\text{air}} gh R^2 \cdot h}{\rho_{\text{water}} R^3 gh} \sim \frac{\rho_{\text{air}}}{\rho_{\text{water}}} \frac{h}{R}$$

Drag dominates when this ratio approaches 1. Given $\rho_{\text{air}}/\rho_{\text{water}} \sim 10^{-3}$ and $h \sim 1$ m, the critical size is $R \sim 1$ mm. Animals smaller than a millimeter lose most of their jumping energy to air friction.

3.3 Heat Conduction in a Compost Pile

Consider a pile of compost (or a biological organism) that generates heat internally due to decomposition. We wish to know how the central temperature scales with the radius of the pile, R .

Let ϵ be the power generated per unit volume. The total heat generated inside the sphere is:

$$P_{\text{gen}} \sim \epsilon R^3$$

To maintain a steady state, this heat must be conducted outwards. According to Fourier's law, the heat flux q (power per unit area) is proportional to the temperature gradient:

$$q = -k \nabla T \sim k \frac{\Delta T}{R}$$

where k is the thermal conductivity and $\Delta T = T_{\text{center}} - T_{\text{surface}}$. The total power lost through the surface area $A \sim R^2$ is:

$$P_{\text{loss}} \sim q R^2 \sim k \frac{\Delta T}{R} R^2 = k R \Delta T$$

Equating generation and loss ($P_{\text{gen}} = P_{\text{loss}}$):

$$\epsilon R^3 \sim k R \Delta T \implies \Delta T \sim \frac{\epsilon}{k} R^2$$

Result: The temperature difference scales as the *square* of the radius. If you double the size of a compost pile, the temperature at the center increases by a factor of 4. This scaling law is crucial for understanding spontaneous combustion in large piles of biological or flammable material.

3.4 Introduction to Dimensional Analysis

We conclude by previewing a formal tool for scaling: Dimensional Analysis (the Buckingham π theorem). By requiring equations to be dimensionally consistent, we can deduce the form of physical laws up to dimensionless constants.

Worked Example 3.1 (The Simple Pendulum)

Consider a pendulum of length L and mass m , in a gravitational field g . We want to find the angle $\theta(t)$. The relevant variables and their dimensions are: $[L] = \text{m}$, $[g] = \text{m/s}^2$, $[m] = \text{kg}$, and $[\theta] = 1$ (dimensionless). Time is $[t] = \text{s}$. Because m is the only parameter with a unit of mass, it cannot combine with anything else to form a dimensionless quantity. Therefore, the pendulum's motion is *independent of its mass*. To form a dimensionless time parameter, we must combine t , g , and L . The only combination yielding seconds is $\sqrt{L/g}$. Thus, the solution must take the form:

$$\theta(t) = f\left(t\sqrt{\frac{g}{L}}\right)$$

implying the period is $T \propto \sqrt{L/g}$.

Worked Example 3.2 (Aerodynamic Drag)

We want to find the drag force F_d on an object of size R moving at velocity v in a fluid of density ρ . Dimensions: $[F_d] = \text{kg} \cdot \text{m/s}^2$, $[\rho] = \text{kg/m}^3$, $[v] = \text{m/s}$, $[R] = \text{m}$. To get kg , we must multiply by ρ . This gives us kg/m^3 . We need m/s^2 , so we multiply by v^2 (giving m^2/s^2). Now we have $\text{kg}/(\text{m} \cdot \text{s}^2)$. Multiplying by R^2 gives the correct units for force:

$$F_d \propto \rho v^2 R^2$$

Worked Example 3.3 (Deep Water Waves)

What is the dispersion relation (frequency ω as a function of wavenumber $k = 2\pi/\lambda$) for deep water waves? The only relevant physical parameters are gravity g and the wavenumber k . (Density ρ drops out as there is no other mass term). Dimensions: $[\omega] = \text{s}^{-1}$, $[g] = \text{m/s}^2$, $[k] = \text{m}^{-1}$. To match the s^{-2} in g , we must use ω^2 . We have ω^2/g with units m^{-1} . Thus, $\omega^2/g \propto k$, yielding:

$$\omega \propto \sqrt{gk}$$

If the water has a finite depth H , we can form a dimensionless parameter kH , and the relation generalizes to $\omega^2 = gkf(kH)$, which must be solved with fluid dynamics.

4 Lecture 4: Dimensional Analysis — From Quantum Atoms to Radiation and Flow

This lecture deepens our practice of dimensional analysis. We move from simple mechanical examples through quantum mechanics (Bohr radius, positronium) to electromagnetism (dipole radiation, Larmor formula) and fluid dynamics (drag force, Reynolds number). A central theme is the power of *identifying which physical parameters are relevant* before building a dimensionless combination.

4.1 Review: Simple Dimensional Analysis Examples

The Physical Picture: In the previous lecture we introduced dimensional analysis. Here we warm up by revisiting three examples that share the same structure: a single length scale, a single acceleration, no mass — so there is a unique dimensionless combination.

The Simple Pendulum

The relevant parameters are length L and gravitational acceleration g . Their dimensions are $[L] = \text{m}$ and $[g] = \text{m s}^{-2}$. We want a time T .

The only combination that yields seconds is:

$$T \propto \sqrt{\frac{L}{g}},$$

so the period is $T = C\sqrt{L/g}$ for some dimensionless constant C (which turns out to be 2π from solving the equation of motion).

Free Fall from Height H

A body dropped from height H in gravity g . The relevant parameters are exactly H and g , so:

$$t_{\text{fall}} \sim \sqrt{\frac{H}{g}}.$$

This is structurally identical to the pendulum — same dimensional reasoning, different physics.

Gravitational Free-Fall Timescale of a Star

A star of mass M and radius r collapses under self-gravity. Now we have three parameters: M , r , and Newton's constant G .

Dimensional check:

$$\begin{aligned} [G] &= \text{kg}^{-1} \text{m}^3 \text{s}^{-2} \\ [M] &= \text{kg}, \quad [r] = \text{m}. \end{aligned}$$

To build a time we note that $[GM] = \text{m}^3 \text{s}^{-2}$, so GM/r^3 has dimensions s^{-2} , giving:

$$t_{\text{ff}} \sim \sqrt{\frac{r^3}{GM}} = \frac{1}{\sqrt{G\bar{\rho}}}$$

where $\bar{\rho} = M/r^3$ is the mean density (up to a numerical factor of order unity).

Physical intuition: The free-fall time depends only on the mean density, not on the actual size or mass separately. A star with the same density as water would fall on a timescale of hours; a white dwarf collapses in seconds.

4.2 The Bohr Radius from Dimensional Analysis

The Physical Picture: The hydrogen atom is a quantum-mechanical bound state of a proton and an electron. The relevant physics is electromagnetic and quantum — there is no classical length scale for the orbit. Dimensional analysis pinpoints the single combination that sets the atomic size.

Relevant parameters:

- Electron charge e (in Gaussian units). Instead of e alone, it is convenient to use e^2 whose dimensions are $[e^2] = \text{erg cm} = \text{g cm}^3 \text{s}^{-2}$.
- Electron mass m_e : $[m_e] = \text{g}$.
- Reduced Planck constant $\hbar$: $[\hbar] = \text{erg s} = \text{g cm}^2 \text{s}^{-1}$.

Why not the speed of light? The hydrogen ground state is non-relativistic (the fine-structure constant $\alpha = e^2/\hbar c \approx 1/137 \ll 1$), so c does not set the scale and we exclude it.

Dimensional construction: We want $[a_0] = \text{cm}$. We must eliminate grams and seconds.

1. *Eliminate seconds.* $\hbar^2$ has dimensions $\text{g}^2 \text{cm}^4 \text{s}^{-2}$, and e^2 has dimensions $\text{g cm}^3 \text{s}^{-2}$. The ratio $\hbar^2/e^2$ has dimensions g cm — one power of mass and one of length remain.
2. *Eliminate grams.* Dividing by m_e removes the gram:

$$\frac{\hbar^2}{m_e e^2} \Rightarrow \left[\frac{\hbar^2}{m_e e^2} \right] = \text{cm}.$$

This is the only combination with units of length, so:

$$a_0 = \frac{\hbar^2}{m_e e^2} \approx 0.529 \text{ \AA} \quad (1)$$

Physical intuition: The Bohr radius is set by the competition between the kinetic energy $\sim \hbar^2/(m_e r^2)$ (from the uncertainty principle, $\Delta p \sim \hbar/r$) and the Coulomb potential energy $\sim e^2/r$. Minimising their sum gives exactly $r = a_0$.

4.3 Hydrogen-Like Atoms: Scaling with Nuclear Charge Z

For an atom with nuclear charge Ze and a single electron (hydrogen-like ion), the only change to the dimensional analysis is $e^2 \rightarrow Ze^2$. Therefore:

$$a_0(Z) = \frac{\hbar^2}{m_e (Ze^2)/Z} = \frac{\hbar^2}{m_e e^2 Z} = \frac{a_0}{Z}. \quad (2)$$

The Bohr radius scales as $1/Z$. The ground-state energy scales as:

$$E_Z = Z^2 E_H = Z^2 \times (-13.6 \text{ eV}).$$

Worked Example 4.1 (Highly Ionised Iron: Fe^{+25})

Iron has $Z = 26$; Fe^{+25} is iron stripped of 25 electrons, leaving one (the innermost). Treating it as hydrogen-like:

$$a_0(\text{Fe}^{+25}) = \frac{a_0}{26} \approx \frac{0.529 \text{ \AA}}{26} \approx 0.020 \text{ \AA}.$$

The binding energy is:

$$E(\text{Fe}^{+25}) = 26^2 \times 13.6 \text{ eV} = 676 \times 13.6 \text{ eV} \approx 9.2 \text{ keV}.$$

This is an X-ray transition, as expected for inner-shell electrons of heavy elements.

4.4 Positronium: A Two-Body Bound State

The Physical Picture: Positronium (Ps) is a bound state of an electron (e^-) and a positron (e^+). There is no heavy nucleus; both particles are equally light. The two-body problem reduces to a one-body problem with the *reduced mass*.

Reduced mass:

$$\mu = \frac{m_e \cdot m_e}{m_e + m_e} = \frac{m_e}{2}.$$

Replacing m_e by μ in Eq. (1):

$$a_0(\text{Ps}) = \frac{\hbar^2}{\mu e^2} = \frac{\hbar^2}{(m_e/2) e^2} = 2a_0 \approx 1.06 \text{ \AA}. \quad (3)$$

Positronium is *twice* as large as hydrogen. The binding energy is correspondingly halved:

$$E(\text{Ps}) = \frac{\mu}{m_e} E_H = \frac{1}{2} (-13.6 \text{ eV}) = -6.8 \text{ eV}.$$

Decay of positronium: Positronium is unstable — the electron and positron annihilate. The number of photons in the final state is fixed by conservation laws. Two cases arise:

- **Para-positronium** (spins anti-parallel, $S = 0$): decays to **2 photons**. Energy conservation requires each photon to carry $m_e c^2 = 511 \text{ keV}$, giving two discrete lines in the spectrum.
- **Ortho-positronium** (spins parallel, $S = 1$): charge-conjugation symmetry forbids 2-photon decay; decays to **3 photons** with a continuous spectrum summing to $2m_e c^2$.

Physical intuition: Para-positronium decays with a lifetime of $\sim 125 \text{ ps}$; ortho-positronium lives $\sim 142 \text{ ns}$ — a factor ~ 1000 longer because it must radiate three photons instead of two.

4.5 Dipole Radiation: Deriving the Larmor Formula

The Physical Picture: An oscillating electric dipole radiates electromagnetic energy. The electric dipole moment is $p = qd$, where q is the charge and d the separation. When

p changes in time (e.g., an accelerating charge), energy is radiated to infinity. We want the radiated power P from dimensional analysis alone.

Relevant parameters: We work in Gaussian units throughout.

- Dipole moment p : $[p] = \text{esu cm} = \text{g}^{1/2} \text{cm}^{3/2} \text{s}^{-1}$. Equivalently, $[p^2] = \text{erg cm}^3$.
- The characteristic oscillation frequency ω : $[\omega] = \text{s}^{-1}$.
- The speed of light c : $[c] = \text{cm s}^{-1}$.

What parameters to exclude: The charge q and distance d appear only in the combination $p = qd$. The medium is vacuum ($\epsilon_0 = 1$ in Gaussian units). No gravitational constant enters.

Dimensional construction: We want $[P] = \text{erg s}^{-1} = \text{g cm}^2 \text{s}^{-3}$.

1. Start with p^2 : dimensions $\text{g cm}^3 \text{s}^{-2}$ (energy $\times$ length).
2. Multiply by ω^4 to introduce extra s^{-4} : $[p^2 \omega^4] = \text{g cm}^3 \text{s}^{-6}$.
3. Divide by c^3 : $[c^3] = \text{cm}^3 \text{s}^{-3}$, so

$$\left[\frac{p^2 \omega^4}{c^3} \right] = \frac{\text{g cm}^3 \text{s}^{-6}}{\text{cm}^3 \text{s}^{-3}} = \text{g s}^{-3} = \text{erg cm}^{-2} \text{s}^{-1}.$$

That's an intensity (power per area), not power — one factor of cm^2 is missing. We need the total power, so we still need an area $\sim (c/\omega)^2 = \lambda^2/(4\pi^2)$ — but that is a dimensionless-factor issue. Including it:

$$P \sim \frac{p^2 \omega^4}{c^3}.$$

Physical insight on the ω^4 dependence: Note that $p\omega^2 \equiv \ddot{p}$ (the second time derivative of the dipole moment). Therefore:

$$\boxed{P \sim \frac{\ddot{p}^2}{c^3}} \quad (4)$$

This is the **Larmor radiation formula** (up to a factor $2/3$ in SI units or $2/(3c^3)$ in Gaussian). Because $\ddot{p} = q\ddot{d} = qa$ for a single charge, the Larmor formula states:

$$P = \frac{2e^2 a^2}{3c^3} \quad (\text{Gaussian}).$$

Why is $P \propto \omega^4$? Each factor of ω comes from a time derivative: one from $\dot{p}$, one from $\ddot{p}$. Squaring gives ω^4 . This is the reason blue light scatters more than red in Rayleigh scattering (λ^{-4} dependence).

Why must P be even in ω (and in q)? By time-reversal symmetry (or charge-conjugation symmetry), the radiated power cannot change sign when we reverse the oscillation direction or flip all charges: hence $P \propto \ddot{p}^2$, not $\ddot{p}$.

4.6 Monopole Radiation is Forbidden

The Physical Picture: Could a time-varying monopole moment $Q(t)$ radiate? If it could, the formula would be $P \sim \ddot{Q}^2/c$ (by dimensional analysis with one fewer power of length). However, **charge conservation** forbids this.

Argument: Gauss's law in vacuum: the total charge $Q = \oint \mathbf{E} \cdot d\mathbf{A}$ is fixed by the electric field on a distant surface. If charge is conserved in the volume, $\dot{Q} = 0$, so $\ddot{Q} = 0$ identically. There is no source for monopole radiation.

Alternative viewpoint (multipole expansion): A single moving charge can always be decomposed, at large distances, as:

$$\text{field} = \underbrace{\text{monopole (static)}}_{\text{no radiation}} + \underbrace{\text{dipole (time-varying)}}_{\text{radiates}},$$

because any motion of the charge creates a time-varying dipole moment $p = q\mathbf{r}(t)$ while the monopole q stays constant. The leading radiated term is therefore *always* dipole, giving Eq. (4).

4.7 Gravitational Wave Radiation: Why It Is So Weak

Analogy with electromagnetism: In electromagnetism the hierarchy is:

$$\text{monopole (forbidden)} \rightarrow \text{dipole (leading)} \rightarrow \text{quadrupole (subleading)}.$$

In gravity, by analogy, we might expect monopole and dipole radiation. Both are forbidden:

- **Monopole forbidden:** mass-energy conservation means $\ddot{M} = 0$.
- **Dipole forbidden:** conservation of momentum means the dipole moment $\mathbf{D} = \sum m_i \mathbf{r}_i$ has $\ddot{\mathbf{D}} = \sum m_i \ddot{\mathbf{r}}_i = 0$ (Newton's third law / center-of-mass motion).

Therefore the **leading gravitational radiation is quadrupolar**:

$$P_{\text{GW}} \sim \frac{G}{c^5} (\ddot{Q}_{ij})^2 \quad (5)$$

where $Q_{ij} = \sum m_a r_i^{(a)} r_j^{(a)}$ is the mass quadrupole tensor. Two extra derivatives (relative to the dipole formula) mean two extra powers of ω , suppressing the radiation further.

Why gravitational waves are so hard to detect: Compared to electromagnetic radiation:

1. Radiation is suppressed by two extra powers of ω (quadrupole vs. dipole), so ordinary astrophysical sources radiate gravitational waves at a tiny rate.
2. Gravitational wave detectors (LIGO/Virgo) couple to the wave via G_N , which is intrinsically tiny, making the transducer inefficient.

Only extremely compact, rapidly-orbiting systems (binary black holes or neutron stars) produce detectable waves. The strain measured by LIGO is of order $h \sim 10^{-21}$, meaning the 4 km arms change length by $\sim 10^{-18}$ m — less than a proton radius.

4.8 Drag Force and the Reynolds Number

The Physical Picture: A solid object of cross-sectional area A moves at velocity v through a fluid of density ρ and dynamic viscosity η . The drag force F_d depends on the regime of flow — turbulent or laminar — which is characterised by a single dimensionless number.

Turbulent (High Reynolds Number) Limit

In the turbulent limit, viscosity is irrelevant to the *bulk* drag (even though it controls the small-scale dissipation). The relevant parameters are ρ , v , and A :

$$[F_d] = \text{g cm s}^{-2}, \quad [\rho] = \text{g cm}^{-3}, \quad [v] = \text{cm s}^{-1}, \quad [A] = \text{cm}^2.$$

The unique combination is:

$$\boxed{F_d = C_D \rho v^2 A} \quad (6)$$

where C_D is a dimensionless drag coefficient of order unity (depends on shape).

The Reynolds Number

To determine whether the flow is turbulent or laminar, we form a dimensionless number from all the relevant parameters. Including viscosity η with $[\eta] = \text{g cm}^{-1} \text{s}^{-1}$, the combination

$$\nu = \frac{\eta}{\rho}, \quad [\nu] = \text{cm}^2 \text{s}^{-1},$$

is the kinematic viscosity. The single dimensionless parameter is:

$$\boxed{\text{Re} = \frac{\sqrt{A} v}{\nu} = \frac{\rho \sqrt{A} v}{\eta}} \quad (7)$$

(using $\sqrt{A}$ as the characteristic length scale of the body).

- **Re $\gg 1$ (turbulent):** The inertia of the fluid dominates; energy cascades from large eddies to progressively smaller ones (Kolmogorov cascade) until it is dissipated at the viscous scale. Drag is given by Eq. (6).
- **Re $\ll 1$ (laminar / Stokes flow):** Viscosity dominates; streamlines are smooth and symmetric.

Laminar (Stokes) Drag

In the laminar limit, the drag force is linear in velocity (from the Navier–Stokes equation linearised in v):

$$F_d \propto \eta v \sqrt{A} \propto \eta v R.$$

For a sphere of radius R this gives the exact Stokes result:

$$F_d = 6\pi \eta v R. \quad (8)$$

How to see this from dimensional analysis: The general result in the laminar limit must take the form

$$F_d = C_D(\text{Re}) \rho v^2 A, \quad \text{with } C_D \propto \text{Re}^{-1} \text{ as } \text{Re} \rightarrow 0.$$

Substituting $C_D = K/\text{Re}$ and $\text{Re} = \rho v \sqrt{A}/\eta$:

$$F_d = \frac{K}{\rho v \sqrt{A}/\eta} \rho v^2 A = K \eta v \sqrt{A}.$$

The factor K is determined by the geometry; for a sphere $K = 6\pi$.

Physical intuition: In laminar flow, the energy is dissipated by viscous shear in the boundary layer. The momentum transferred per unit time (= drag force) is proportional to the velocity gradient, which is $\sim v/R$, times the area R^2 and the viscosity: $F \sim \eta(v/R)R^2 = \eta v R$.

Summary of Key Results

System	Result	Key parameters
Stellar free-fall time	$t_{\text{ff}} \sim 1/\sqrt{G\rho}$	G, M, r
Bohr radius	$a_0 = \hbar^2/(m_e e^2)$	$\hbar, m_e, e$
Positronium radius	$a_0(\text{Ps}) = 2a_0$	reduced mass $\mu = m_e/2$
Hydrogen-like radius	$a_0(Z) = a_0/Z$	nuclear charge Z
Larmor radiation	$P \sim \ddot{p}^2/c^3$	p, ω, c
GW emission	$P_{\text{GW}} \sim G\ddot{Q}^2/c^5$	quadrupole, suppressed
Turbulent drag	$F_d \sim \rho v^2 A$	ρ, v, A
Laminar drag (Stokes)	$F_d = 6\pi\eta v R$	η, v, R
Reynolds number	$\text{Re} = \rho v \sqrt{A}/\eta$	determines flow regime

Recommended Reading

- **Dimensional Analysis (general):** Barenblatt, *Scaling, Self-Similarity, and Intermediate Asymptotics*, Ch. 1–3 — the definitive treatment; covers the Buckingham π theorem rigorously and applies it to fluid mechanics and beyond.
- **Bohr radius and hydrogen atom:** Griffiths, *Introduction to Quantum Mechanics* (3rd ed.), §4.2 — derives the energy levels and wavefunctions from first principles, making explicit the role of a_0 .
- **Larmor formula and dipole radiation:** Griffiths, *Introduction to Electrodynamics* (4th ed.), §11.2 — clear derivation of the Larmor formula and its generalisations. Jackson, *Classical Electrodynamics* (3rd ed.), Ch. 9 — the authoritative reference for multipole radiation.
- **Reynolds number and fluid flow:** Batchelor, *An Introduction to Fluid Dynamics*, §4.9 — derivation of the Reynolds number from the Navier–Stokes equation and discussion of the laminar-to-turbulent transition.
- **Gravitational waves:** Maggiore, *Gravitational Waves: Theory and Experiments* (Oxford), Ch. 1 — excellent introduction to the quadrupole formula and why monopole/dipole radiation are absent. [LIGO Scientific Collaboration, GW150914 \(2016\)](#) — the landmark detection paper.

- **Positronium:** [Wikipedia: Positronium](#) — good overview of the two states, their lifetimes, and decay modes.

5 Lecture 5: Buckingham's Pi Theorem, Blast Waves, and Rocket-Movie Geometry

This lecture turns dimensional analysis from a collection of examples into a general theorem. Buckingham's Pi theorem tells us how many independent dimensionless variables are needed to describe a physical problem. We then use that theorem in two observational problems: estimating the energy of a port explosion from video frames, and testing whether a visible expanding feature in a missile-interception movie can really be interpreted as a blast wave.

The central lesson is methodological. Dimensional analysis does not only give quick formulas. It also tells us how many parameters an experiment must vary, which variables can be irrelevant, and which interpretation of an observation is internally consistent.

5.1 Buckingham's Pi Theorem

The Physical Picture: Physical laws cannot depend on our arbitrary choice of units. A relation between measured quantities must therefore look the same if we switch from meters to centimeters, seconds to hours, or kilograms to grams. Buckingham's Pi theorem is the systematic way to convert this unit-invariance into a count of the truly independent, dimensionless variables.

General relation

Suppose a physical problem contains N dimensional variables $x_1, \dots, x_N$. A completely general relation among them can be written as

$$f(x_1, x_2, \dots, x_N) = 0. \quad (9)$$

Assume that the variables use M independent base units $u_1, \dots, u_M$. Each variable can be decomposed into a numerical value v_i times powers of the units:

$$x_i = v_i \prod_{j=1}^M u_j^{\alpha_{ij}}, \quad (10)$$

where α_{ij} is the power of unit u_j in x_i . For example, if x_i is a velocity and the base units are length and time, then $\alpha_{iL} = 1$ and $\alpha_{iT} = -1$.

Unit transformations

Now rescale the base units by

$$u_j \longrightarrow u_j^{\text{new}} = \lambda_j u_j. \quad (11)$$

The physical relation must remain true:

$$f(x_1^{\text{new}}, x_2^{\text{new}}, \dots, x_N^{\text{new}}) = 0. \quad (12)$$

To extract the consequence of this invariance, differentiate with respect to one scale factor λ_k and then evaluate at $\lambda_j = 1$ for all j . By the chain rule,

$$\begin{aligned} 0 &= \left. \frac{\partial}{\partial \lambda_k} f(x_1^{\text{new}}, \dots, x_N^{\text{new}}) \right|_{\lambda_j=1} \\ &= \sum_{i=1}^N \left. \frac{\partial f}{\partial x_i} \frac{\partial x_i^{\text{new}}}{\partial \lambda_k} \right|_{\lambda_j=1}. \end{aligned} \quad (13)$$

Since

$$x_i^{\text{new}} = v_i \prod_{j=1}^M (\lambda_j u_j)^{\alpha_{ij}},$$

differentiating with respect to λ_k gives

$$\left. \frac{\partial x_i^{\text{new}}}{\partial \lambda_k} \right|_{\lambda_j=1} = \alpha_{ik} x_i.$$

Therefore each independent unit gives one differential constraint:

$$\boxed{\sum_{i=1}^N \alpha_{ik} x_i \frac{\partial f}{\partial x_i} = 0, \quad k = 1, \dots, M.} \quad (14)$$

Physical significance: Each independent unit-rescaling removes one independent dimensional direction from the problem.

The theorem

We began with one relation among N variables. Unit invariance adds M independent constraints, provided the M units are genuinely independent in the variables of the problem. Thus the physical content can be described using $N - M$ independent dimensionless combinations:

$$\boxed{F(\Pi_1, \Pi_2, \dots, \Pi_{N-M}) = 0.} \quad (15)$$

Physical significance: Instead of exploring an N -dimensional parameter space, the experiment or theory only needs the $N - M$ dimensionless directions.

The Π_i are not unique. If Π_1 and Π_2 are valid dimensionless groups, then $\Pi_1 \Pi_2$, Π_1 / Π_2 , or any non-singular redefinition can also be used. A useful choice puts the quantity we want to solve for in only one dimensionless group, so that it can be isolated after the dimensionless relation is written.

5.2 Example: Deflection in a Falling Body

The Physical Picture: If a body is dropped from a high tower, Earth's rotation produces a small Coriolis deflection. We may not know the exact coefficient without solving the equations of motion, but dimensional analysis already tells us the possible parameter dependence.

Let Δx be the horizontal deflection, H the height, g the gravitational acceleration, Ω Earth's angular velocity, $R_\oplus$ Earth's radius, and m the body's mass. The variables use the

dimensions M, L, T , so $N = 6$ and $M = 3$. We expect three independent dimensionless groups. A convenient choice is

$$\Pi_1 = \frac{\Delta x}{H}, \quad \Pi_2 = \frac{H}{R_\oplus}, \quad \Pi_3 = \frac{g}{\Omega^2 R_\oplus}. \quad (16)$$

The mass cannot enter any nontrivial dimensionless group because no other variable carries units of mass. Thus the most general dimensionless relation can be written as

$$F\left(\frac{\Delta x}{H}, \frac{H}{R_\oplus}, \frac{g}{\Omega^2 R_\oplus}\right) = 0. \quad (17)$$

Equivalently, after isolating Δx ,

$$\Delta x = H \Phi\left(\frac{H}{R_\oplus}, \frac{g}{\Omega^2 R_\oplus}\right). \quad (18)$$

Physical significance: Dimensional analysis proves immediately that the dropped body's mass is irrelevant and that only two dimensionless control parameters remain.

Physics gives one more piece of information. The Coriolis acceleration is linear in Ω and in the falling velocity. In the regime $H \ll R_\oplus$, the dependence must therefore reduce to

$$\Delta x \sim C \Omega \frac{H^{3/2}}{g^{1/2}}, \quad (19)$$

where C is a dimensionless coefficient, including the latitude dependence. *Physical significance:* The theorem gives the allowed dimensionless form; the equation of motion supplies the leading power of the small rotational effect.

5.3 Strong Blast Waves

The Physical Picture: A strong explosion releases energy into a small region. At early times the pressure and energy density inside the blast are much larger than the ambient air pressure, so the unperturbed pressure can be neglected. The blast front then expands supersonically, and the question is how its radius $r(t)$ depends on time, explosion energy, and air density.

Relevant variables

For a strong spherical blast in a uniform medium, take

- r : radius of the shock front,
- t : time since the explosion,
- ρ : ambient density,
- E : total released energy that drives the blast.

Their dimensions are

$$[r] = L, \quad [t] = T, \quad [\rho] = ML^{-3}, \quad [E] = ML^2T^{-2}.$$

There are $N = 4$ variables and $M = 3$ independent units, so Buckingham's theorem says there is one dimensionless group.

Let

$$\Pi = r^w t^x E^y \rho^z.$$

Demanding zero net powers of L, T, M gives

$$\begin{aligned} L : \quad w + 2y - 3z &= 0, \\ T : \quad x - 2y &= 0, \\ M : \quad y + z &= 0. \end{aligned} \tag{20}$$

Keeping w free, the solution is

$$x = -\frac{2w}{5}, \quad y = -\frac{w}{5}, \quad z = \frac{w}{5}. \tag{21}$$

Choosing $w = 1$ gives

$$\Pi = r t^{-2/5} E^{-1/5} \rho^{1/5}. \tag{22}$$

Therefore the radius must have the Sedov-Taylor scaling

$$\boxed{r(t) = \Pi' \left(\frac{E}{\rho} \right)^{1/5} t^{2/5}.} \tag{23}$$

Physical significance: A stronger explosion expands farther, but only as $E^{1/5}$; the fifth root makes blast-radius estimates relatively insensitive to the exact energy.

Solving the same relation for the energy gives

$$\boxed{E = \Pi'' \rho \frac{r^5}{t^2}.} \tag{24}$$

Physical significance: Video measurements of r and t can estimate the explosion energy, provided the observed front is still a strong shock.

Calibration from a known explosion

Dimensional analysis does not determine the dimensionless coefficient Π'' . One way to estimate it is to compare with a known explosion. Using Trinity as an order-of-magnitude calibration, with $E \sim 10^{14}$ J, $r \sim 130$ m, $t \sim 0.025$ s, and $\rho \sim 1.2 \text{ kg m}^{-3}$,

$$\boxed{\Pi'' \simeq E r^{-5} t^2 \rho^{-1} \sim 1.4.} \tag{25}$$

Physical significance: The calibrated coefficient lets us turn the scaling law into a practical estimator, while still keeping only order-unity precision.

5.4 Bandar Abbas Port Explosion

The Physical Picture: On April 26, 2025, a large explosion occurred at the Shahid Rajaei port near Bandar Abbas, Iran. Public videos show a visible blast front. The question is whether the frames are close enough to the explosion for the strong-shock formula to estimate the released energy.

Camera far from the explosion

In one video, the visible front travels about 425 m in 45 frames at 30 frames per second. The average speed is therefore

$$v_{\text{avg}} \simeq 425 \text{ m} \frac{30 \text{ s}^{-1}}{45} \simeq 300 \text{ m s}^{-1}. \quad (26)$$

This is already comparable to the speed of sound, so the shock is weak. In that regime its speed is controlled mainly by the acoustic speed rather than by the original explosion energy. This video therefore gives a poor energy estimate.

Camera closer to the explosion

A closer camera gives a displacement of about 69 m in 4 frames at 30 frames per second:

$$v_{\text{avg}} \simeq 69 \text{ m} \frac{30 \text{ s}^{-1}}{4} \simeq 5.2 \times 10^2 \text{ m s}^{-1}. \quad (27)$$

This is significantly above the speed of sound, so the strong-shock approximation is more plausible. Using Eq. (24) with $r \simeq 69 \text{ m}$, $t = 4/30 \text{ s}$, $\rho \simeq 1.2 \text{ kg m}^{-3}$, and $\Pi'' \simeq 1.4$,

$$E \simeq 1.4 (1.2 \text{ kg m}^{-3}) \frac{(69 \text{ m})^5}{(4/30 \text{ s})^2} \simeq 1.5 \times 10^{11} \text{ J}. \quad (28)$$

Since 1 ton of TNT corresponds to roughly $4.2 \times 10^9 \text{ J}$,

$$\boxed{E \sim 35 \text{ ton TNT}, \quad \text{with a plausible range} \sim 20\text{--}55 \text{ ton TNT}.} \quad (29)$$

Physical significance: The result is not a precise forensic measurement; it is an order-of-magnitude energy estimate controlled mainly by frame timing and distance calibration.

The sensitivity to measurement errors is easy to understand from $E \propto r^5 t^{-2}$. A moderate fractional error in radius is amplified by a factor of five in the energy, while a one-frame error in time also matters strongly when only four frames are used.

5.5 Rocket-Movie Geometry and a Consistency Check

The Physical Picture: The lecture then considered a movie of a missile interception seen from Mevasseret Zion shortly after sunset. The visible expanding feature is bright against the dark sky because it appears to be sunlit. Its angular radius can be measured frame by frame. The first tempting interpretation is a blast wave, but dimensional analysis and geometry allow us to test that interpretation.

Weak-blast interpretation

The measured angular radius is approximately linear:

$$\theta(t) \simeq \theta_0 + \dot{\theta} t, \quad \dot{\theta} \simeq 1.0046 \times 10^{-2} \text{ rad s}^{-1}. \quad (30)$$

If the feature is already a weak blast wave, its physical expansion speed is approximately the local sound speed. Taking $c_s \simeq 310 \text{ m s}^{-1}$,

$$\boxed{D \simeq \frac{c_s}{\dot{\theta}} \simeq \frac{310}{1.0046 \times 10^{-2}} \text{ m} \simeq 31 \text{ km}.} \quad (31)$$

Physical significance: If the observed edge moves at the sound speed, the angular expansion directly gives the distance to the event.

For an altitude angle $\alpha \simeq 35^\circ$, this would imply

$$h \simeq D \sin \alpha \sim 18 \text{ km} \quad \text{or, using horizontal geometry, } h \sim 22 \text{ km.} \quad (32)$$

This is a low-altitude answer for a ballistic-missile interception, but not yet a contradiction by itself.

Energy upper bound from early linearity

If the blast wave is already weak in the first measured frame, then the Sedov-Taylor strong-shock velocity must have fallen to the sound speed before or around that radius. From Eq. (23),

$$v = \frac{dr}{dt} = \frac{2r}{5t}. \quad (33)$$

Setting $v \sim c_s$ gives $t \sim 2r/(5c_s)$. Substituting this into Eq. (24) gives

$$E \lesssim \frac{25}{4} \Pi'' \rho c_s^2 r_1^3. \quad (34)$$

For $\theta_1 \simeq 1 \text{ mrad}$ and $D \simeq 31 \text{ km}$, $r_1 \simeq D\theta_1 \simeq 31 \text{ m}$. With $\Pi'' \simeq 1.4$, $\rho \simeq 0.05 \text{ kg m}^{-3}$, and $c_s \simeq 310 \text{ m s}^{-1}$,

$$E \lesssim 1.3 \times 10^9 \text{ J} \simeq 0.3 \text{ ton TNT}. \quad (35)$$

Physical significance: Under the weak-blast interpretation, the observed early linearity gives only an upper bound on the explosive energy.

The sunlight test

The key extra observation is that the feature is still illuminated by the Sun after local sunset. If the Sun is δ below the horizon, a point must be high enough above Earth's shadow to receive direct sunlight. The geometric condition is

$$h \gtrsim R_\oplus (\sec \delta - 1). \quad (36)$$

Physical significance: Illumination provides a distance and height constraint that is independent of the blast-wave dynamics.

For the movie, $\delta \simeq 7.6^\circ$, so

$$h \gtrsim 6371 \text{ km} (\sec 7.6^\circ - 1) \simeq 57 \text{ km.} \quad (37)$$

At an altitude angle $\alpha \simeq 35^\circ$, the slant distance is therefore

$$D \gtrsim \frac{h}{\sin \alpha} \simeq \frac{57 \text{ km}}{\sin 35^\circ} \sim 100 \text{ km.} \quad (38)$$

Physical significance: The sunlit geometry pushes the event much farther away than the weak-blast distance estimate.

If the measured angular radius reaches $\theta \simeq 3 \text{ mrad}$, then the physical radius is at least

$$r \gtrsim D\theta \sim (100 \text{ km})(3 \times 10^{-3}) \sim 300 \text{ m.} \quad (39)$$

Swept-up mass argument

At $h \sim 57$ km, a representative atmospheric density is $\rho_{\text{air}} \sim 3 \times 10^{-4} \text{ kg m}^{-3}$. The mass of air inside a sphere of radius $r \sim 300$ m is

$$M_{\text{air}} \sim \frac{4\pi}{3} \rho_{\text{air}} r^3 \sim \frac{4\pi}{3} (3 \times 10^{-4}) (300)^3 \text{ kg} \sim 3 \times 10^4 \text{ kg}. \quad (40)$$

This is tens of tons of air. A compact vapor cloud or gas shell from the interception could not expand through that much air at almost constant speed unless its own mass and energy were enormous.

The visible feature is probably not a classical blast wave.

(41)

Physical significance: The best interpretation is a sparse cloud of sunlit fragments, dust, smoke, or trail material, whose apparent boundary grows in projection rather than a dense shell sweeping up all the surrounding air.

5.6 Summary of the Method

The Physical Picture: Dimensional analysis is strongest when it is paired with observational checks. It can produce a scaling law, but it also tells us when that scaling law is no longer the right model.

- Buckingham's theorem reduces N dimensional variables with M independent units to $N - M$ dimensionless variables.
- For a strong blast wave, the only possible scaling is $r \propto (E/\rho)^{1/5} t^{2/5}$.
- A video can estimate an explosion energy only if the observed front is still a strong shock.
- In the Bandar Abbas example, a close camera gave a plausible estimate of order tens of tons of TNT.
- In the rocket movie, sunlight geometry contradicted the naive weak-blast distance estimate, pushing the interpretation toward sparse sunlit debris rather than a classical blast wave.

6 Lecture 6: Diffusion, Heat Conduction, Viscosity, and the Reynolds Number

This lecture unifies three apparently different physical phenomena — particle diffusion, heat conduction, and viscous flow — by showing that they all arise from the *same microscopic mechanism*: particles random-walking and transporting whatever scalar or vector quantity they carry. Dimensional analysis then fixes the form of each transport coefficient, and dimensionless ratios of these coefficients encode deep physics about the dominant carrier.

6.1 What Is Diffusion?

The Physical Picture: Diffusion is the net transport of a quantity (particle number, heat, momentum) driven by a gradient, through the statistical bias of random walkers. If there are more ink molecules on one side of an imaginary surface than on the other, more will cross from the dense side per unit time than from the dilute side, producing a net flux even though individual particles move randomly. The same picture applies to heat (particles carry thermal energy) and to momentum (particles carry bulk velocity).

Canonical examples:

- Dye in a liquid: ink diffuses until concentration is uniform.
- Heat: Fourier's law—heat flows from hot to cold.
- Momentum: viscosity—fast fluid layers drag slow ones.
- Social/biological systems: epidemic spreading, nutrient transport.

6.2 Fick's Law from Dimensional Analysis

Physical Motivation

We want to find the particle flux $\mathbf{J}$ (number of particles crossing unit area per unit time) in terms of the fundamental microscopic quantities. The relevant parameters are:

- n — local particle number density, $[n] = \text{m}^{-3}$,
- ℓ — mean free path (typical distance a particle travels before a collision resets its direction), $[\ell] = \text{m}$,
- v — typical thermal speed, $[v] = \text{m s}^{-1}$.

The quantity to find has dimensions $[J] = \text{m}^{-2} \text{s}^{-1}$.

Dimensional counting

Build J from n , ℓ , and v . There are three variables and three independent base dimensions (m, s, and the pure number count embedded in n), so Buckingham's theorem leaves one dimensionless combination,

$$\Pi = n \ell^3, \quad (42)$$

and the most general dimensional relation is

$$J = n v \ell \Phi(n \ell^3), \quad (43)$$

where Φ is an unknown function. The factor $n v \ell$ has units $\text{m}^{-3} \cdot \text{m s}^{-1} \cdot \text{m} = \text{m}^{-2} \text{s}^{-1}$ as required.

Linearity selects the leading term

When the number density is small (dilute limit, $n\ell^3 \ll 1$), particles interact rarely and the flux must be linear in n : doubling the number of particles doubles the flux. This forces $\Phi = \text{const}$, giving

$$J \sim n v \ell. \quad (44)$$

More carefully, the flux is the imbalance between particles crossing a surface from left and right. The particles that reach the surface from the left started roughly one mean free path ℓ away:

$$J_+ - J_- \approx n\left(x - \frac{\ell}{2}\right) \frac{v}{2} - n\left(x + \frac{\ell}{2}\right) \frac{v}{2} = -\frac{v\ell}{2} \frac{\partial n}{\partial x}. \quad (45)$$

Collecting factors of order unity gives Fick's first law:

$$\boxed{J = -D \nabla n, \quad D \sim \ell v.} \quad (46)$$

Physical significance: The diffusion coefficient D has units $\text{m}^2 \text{s}^{-1}$; it sets the rate at which a concentration pattern spreads in space. The minus sign reflects the physical fact that transport is down the gradient.

6.3 Estimating the Diffusion Coefficient in Air

Physical Motivation

To make $D \sim \ell v$ concrete, we need the mean free path ℓ and thermal speed v of air molecules at room temperature.

Thermal speed

The Maxwell-Boltzmann distribution for one component of velocity gives

$$v_{\text{th}} \sim \sqrt{\frac{k_B T}{m}}. \quad (47)$$

For air (average molecular mass $\bar{m} \approx 29 \text{ u} \approx 4.8 \times 10^{-26} \text{ kg}$) at $T = 300 \text{ K}$:

$$v_{\text{th}} \sim \sqrt{\frac{1.38 \times 10^{-23} \times 300}{4.8 \times 10^{-26}}} \simeq 300\text{--}500 \text{ m s}^{-1}. \quad (48)$$

In practice one takes $v_{\text{th}} \simeq 340\text{--}500 \text{ m s}^{-1}$ (speed of sound is the lower bound for a related quantity; the true rms speed is somewhat higher).

Mean free path

A particle sweeps out a cylinder of cross-sectional area σ (the collision cross section). It collides when another particle lies inside this cylinder. The mean free path is therefore

$$\ell = \frac{1}{\sqrt{2} n \sigma}, \quad (49)$$

where the $\sqrt{2}$ accounts for relative velocities in a moving gas. Estimating the cross section from the molecular size $d \sim 3 \text{ \AA} = 3 \times 10^{-10} \text{ m}$:

$$\sigma \sim \pi \left(\frac{d}{2}\right)^2 \approx \pi \times (1.5 \times 10^{-10})^2 \approx 7 \times 10^{-20} \text{ m}^2. \quad (50)$$

The number density of air at sea level follows from $n = P/(k_B T)$ or from the ratio to water:

$$n_{\text{air}} = \frac{P_0}{k_B T} \approx \frac{10^5 \text{ Pa}}{1.38 \times 10^{-23} \text{ J K}^{-1} \times 300 \text{ K}} \approx 2.4 \times 10^{25} \text{ m}^{-3}. \quad (51)$$

Alternatively, water has $n_{\text{water}} = \rho_w N_A / M_{\text{mol}} \approx 3.3 \times 10^{28} \text{ m}^{-3}$, and since $\rho_{\text{air}}/\rho_w \approx 1.3 \times 10^{-3}$, we get $n_{\text{air}} \approx 4 \times 10^{25} \text{ m}^{-3}$ — consistent to within a factor of two. Therefore

$$\ell \approx \frac{1}{\sqrt{2} \times 2.4 \times 10^{25} \times 7 \times 10^{-20}} \approx \frac{1}{\sqrt{2} \times 1.7 \times 10^6 \text{ m}^{-1}} \approx 4 \times 10^{-7} \text{ m} \approx 100 \text{ nm}. \quad (52)$$

(The real value is $\approx 70 \text{ nm}$ at 20°C , consistent with our estimate.) The diffusion coefficient is therefore

$$D_{\text{air}} \sim \ell v_{\text{th}} \approx 10^{-7} \text{ m} \times 5 \times 10^2 \text{ m s}^{-1} \approx 5 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}. \quad (53)$$

Physical significance: The real value for air is $D \approx 2 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$, in good order-of-magnitude agreement. The estimate succeeded using only molecular size and temperature.

6.4 The Diffusion Equation and Characteristic Timescales

Physical Motivation

If particles flow according to Fick's first law, the continuity equation $\partial n / \partial t + \nabla \cdot \mathbf{J} = 0$ immediately gives the diffusion equation (Fick's second law):

$$\frac{\partial n}{\partial t} = D \nabla^2 n. \quad (54)$$

Physical significance: The sign of $\nabla^2 n$ is negative at a concentration peak (divergence of flux is positive there — more leaves than arrives) so the equation says peaks erode and troughs fill, driving the system toward uniformity.

Characteristic timescale by dimensional analysis

The diffusion equation contains only two scales: the diffusion coefficient D ($[\text{m}^2 \text{ s}^{-1}]$) and the length scale L of the initial pattern. The only combination with units of time is

$$\tau_{\text{diff}} \sim \frac{L^2}{D}. \quad (55)$$

Physical significance: The quadratic dependence on L is characteristic of diffusion: doubling the length scale quadruples the spread time. This makes diffusion extremely slow over macroscopic distances.

Example: Perfume in a room

How long does it take for perfume molecules to diffuse across a room of size $L \sim 1 \text{ m}$?

$$\tau_{\text{diff}} \sim \frac{(1 \text{ m})^2}{2 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}} \approx 5 \times 10^4 \text{ s} \approx 14 \text{ hours}. \quad (56)$$

Yet in practice you smell perfume within a second or two. The discrepancy is large: a factor of $\sim 10^4$.

Turbulent diffusion

The resolution is that everyday airflow is *turbulent*. Macroscopic eddies — stirring motions with effective size L_{turb} and turnover speed v_{turb} — mix the gas on timescale $\tau_{\text{turb}} \sim L_{\text{turb}}/v_{\text{turb}}$. Defining an effective turbulent diffusivity:

$$D_{\text{turb}} \sim v_{\text{turb}} L_{\text{turb}}. \quad (57)$$

To smell perfume in $\tau \sim 1$ s across $L \sim 1$ m, we need

$$D_{\text{turb}} \sim \frac{L^2}{\tau} \sim \frac{1 \text{ m}^2}{1 \text{ s}} \approx 1 \text{ m}^2 \text{ s}^{-1}, \quad (58)$$

which is five orders of magnitude larger than the molecular D_{air} . *Physical significance:* Turbulent mixing completely dominates molecular diffusion at human scales in air. In the atmosphere, diffusion is also anisotropic: horizontal eddy diffusion is far more efficient than vertical diffusion because the vertical density gradient strongly resists upward displacement.

6.5 Heat Conduction from the Same Mechanism

Physical Motivation

Each gas molecule carries kinetic energy $\varepsilon \sim k_B T$ (per degree of freedom). When a molecule random-walks from a hot region into a cold one, it deposits energy. The heat flux is therefore the *energy content per particle* times the particle flux.

Derivation of Fourier's law

Let $\varepsilon(T)$ be the internal energy per particle. The heat flux is

$$\mathbf{J}_Q = \varepsilon(T) \mathbf{J} = -D \varepsilon(T) \nabla n. \quad (59)$$

If the number density is uniform but the temperature varies, $\nabla n \approx 0$ but $\nabla(n\varepsilon) = n \nabla \varepsilon = n \frac{d\varepsilon}{dT} \nabla T$. Therefore

$$\mathbf{J}_Q = -D n \frac{d\varepsilon}{dT} \nabla T = -D c_V \nabla T, \quad (60)$$

where $c_V = n d\varepsilon/dT$ is the heat capacity per unit volume. This is Fourier's law with thermal conductivity $\kappa = D c_V$:

$$\mathbf{J}_Q = -\kappa \nabla T, \quad \kappa = D c_V. \quad (61)$$

Physical significance: The thermal conductivity is not an independent property — it is the diffusion coefficient times the heat capacity per volume. For ideal gases, this means $\kappa \sim \ell v n k_B$.

The heat diffusion equation

Applying continuity to the energy density $u = c_V T$:

$$\rho c_p \frac{\partial T}{\partial t} = -\nabla \cdot \mathbf{J}_Q = \kappa \nabla^2 T, \quad (62)$$

which gives

$$\boxed{\frac{\partial T}{\partial t} = D_{\text{thermal}} \nabla^2 T, \quad D_{\text{thermal}} = \frac{\kappa}{\rho c_p}.} \quad (63)$$

Physical significance: The thermal diffusivity D_{thermal} has the same units and the same physical origin as the particle diffusion coefficient. For an ideal monatomic gas, $c_V \sim \frac{3}{2}nk_B$ and $\rho c_p \sim \frac{5}{2}nk_B$, so $D_{\text{thermal}} = \frac{3}{5}D \sim D$. The small numeric factor confirms that heat spreads on the same timescale as particle density for ideal gases.

Note that c_p appeared instead of c_V because work done on adjacent fluid elements must be included (constant-pressure process). The ratio $\gamma = c_p/c_V$ appears, but it is $O(1)$ and does not change the order-of-magnitude conclusion.

6.6 Momentum Diffusion and Viscosity

Physical Motivation

Gas molecules also carry bulk momentum. In a shear flow where the x -velocity v_x depends on y , a molecule crossing from a fast layer at $y - \ell$ to a slow layer at y carries extra x -momentum and deposits it, tending to accelerate the slow layer. This is viscosity — friction from momentum diffusion.

Viscous stress and kinematic viscosity

The momentum flux (stress) in the x -direction through a surface of constant y is, by the same mean-free-path argument:

$$\sigma_{xy} = -\nu \rho \frac{\partial v_x}{\partial y}, \quad \nu \sim \ell v_{\text{th}}, \quad (64)$$

where $\nu = \eta/\rho$ is the kinematic viscosity ($[\nu] = \text{m}^2 \text{s}^{-1}$). The dynamic viscosity is $\eta = \rho \nu$.

$$\boxed{\nu \sim \ell v_{\text{th}} \sim D_{\text{particle}}.} \quad (65)$$

Physical significance: For an ideal gas, kinematic viscosity, particle diffusivity, and thermal diffusivity are all set by the *same* product ℓv_{th} . They differ only by numerical prefactors of order unity.

The Navier-Stokes equation

Applying momentum conservation to a fluid element and adding the viscous diffusion term gives the Navier-Stokes equation:

$$\boxed{\rho \left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} \right) = -\nabla P + \eta \nabla^2 \mathbf{v} + \rho \mathbf{g}.} \quad (66)$$

- The $(\mathbf{v} \cdot \nabla) \mathbf{v}$ term is the *advective* (convective) derivative — it is nonlinear in $\mathbf{v}$ and responsible for turbulence.
- The $\eta \nabla^2 \mathbf{v}$ term is the viscous diffusion of momentum.
- The pressure gradient $-\nabla P$ drives flow from high to low pressure.

Physical significance: The richness of fluid dynamics comes entirely from the nonlinear advection term, which the viscous term opposes. Their ratio is the Reynolds number.

6.7 The Prandtl Number

Physical Motivation

We have three transport coefficients, all with the same units: D (particle diffusivity), D_{thermal} (thermal diffusivity), and ν (kinematic viscosity). For an ideal gas they are all $\sim \ell v_{\text{th}}$. But in a liquid metal or a polymer melt the dominant carriers of heat and momentum are completely different. The Prandtl number encodes this.

$$\Pr = \frac{\nu}{D_{\text{thermal}}} = \frac{\text{kinematic viscosity}}{\text{thermal diffusivity}}. \quad (67)$$

Physical significance: $\Pr$ compares how efficiently momentum versus heat is diffused.

Air ($\Pr \sim 1$)

Both momentum and heat are carried by the same air molecules via the same mean-free-path mechanism. Therefore $\nu \sim D_{\text{thermal}}$ and $\Pr_{\text{air}} \approx 0.71$ — of order unity, as expected.

Mercury ($\Pr \ll 1$)

Mercury is a liquid metal. Its atoms carry momentum through inter-atomic collisions (same mechanism as gases, but in a dense liquid). However, heat is mainly carried by *conduction electrons*, which are far lighter, faster, and more numerous than the atoms. Electrons diffuse heat much more efficiently than the heavy atoms diffuse momentum:

$$D_{\text{thermal}}^{\text{Hg}} \gg \nu^{\text{Hg}} \quad \Rightarrow \quad \Pr_{\text{Hg}} \approx 0.025 \ll 1. \quad (68)$$

The general rule: *anything that conducts electricity well (mobile electrons) will also conduct heat well*, because the electron mean free path is long and their speed is the Fermi velocity ($\sim 10^6 \text{ m s}^{-1}$), not the thermal velocity. This is the Wiedemann-Franz law.

Long-chain polymers ($\Pr \gg 1$)

A tangle of polymer chains transmits momentum very efficiently: pulling one end of a chain exerts a force at the far end along the chain backbone, giving extremely high viscosity. Heat transport, by contrast, still requires molecular collisions and is slow. Consequently

$$\nu^{\text{polymer}} \gg D_{\text{thermal}}^{\text{polymer}} \quad \Rightarrow \quad \Pr_{\text{polymer}} \gg 1. \quad (69)$$

Material	$\Pr$	Dominant heat carrier
Air (ideal gas)	≈ 0.7	Molecules
Water	≈ 7	Molecules (H-bond network)
Mercury	≈ 0.025	Conduction electrons
Polymer melt	$\gg 10^2$	No efficient carrier

6.8 Drag on a Sphere and the Physical Meaning of Reynolds Number

Physical Motivation

A sphere of radius R moves at speed V through a fluid of density ρ and kinematic viscosity ν . We want the drag force F using dimensional analysis, and then connect Re to the competition between two timescales.

Dimensional analysis of drag

The variables are F (force, $[MLT^{-2}]$), ρ ($[ML^{-3}]$), V ($[LT^{-1}]$), R ($[L]$), and ν ($[L^2T^{-1}]$). That is $N = 5$ variables and $M = 3$ independent base dimensions, so there are two dimensionless groups. Choose

$$\Pi_1 = \frac{F}{\rho V^2 R^2}, \quad \Pi_2 = \text{Re} = \frac{VR}{\nu}. \quad (70)$$

Buckingham's theorem then gives

$$\boxed{F = \rho V^2 R^2 f(\text{Re})}, \quad (71)$$

where f is an unknown function that experiment or theory must supply. *Physical significance:* The drag coefficient $C_D \equiv F/(\rho V^2 A)$ is a function of Re alone (for a given body shape). All drag data collapse onto a single curve when plotted as C_D vs. Re .

Stokes regime ($\text{Re} \ll 1$)

At low Reynolds number, the fluid has time to adjust smoothly and viscosity dominates. The problem is linear in V , so the force must be linear in V . This is only consistent with Eq. (71) if

$$f(\text{Re}) \propto \frac{1}{\text{Re}} \Rightarrow F \propto \rho V^2 R^2 \cdot \frac{\nu}{VR} = \rho \nu V R = \eta V R. \quad (72)$$

The exact solution (Stokes, 1851) gives the numerical prefactor:

$$\boxed{F_{\text{Stokes}} = 6\pi \eta V R}. \quad (73)$$

Physical significance: Stokes drag is linear in speed and viscosity, and linear in size — very different from turbulent drag ($F \propto V^2 R^2$). Small organisms (bacteria, pollen) and falling droplets live in the Stokes regime.

Physical interpretation of the Reynolds number

Two natural timescales appear in the problem:

$$\tau_{\text{cross}} = \frac{R}{V} \quad (\text{time for fluid to flow past the object}), \quad (74)$$

$$\tau_{\text{diff}} = \frac{R^2}{\nu} \quad (\text{time for momentum to diffuse a distance } R). \quad (75)$$

Their ratio is exactly the Reynolds number:

$$\boxed{\text{Re} = \frac{\tau_{\text{diff}}}{\tau_{\text{cross}}} = \frac{R^2/\nu}{R/V} = \frac{VR}{\nu}}. \quad (76)$$

Physical significance:

- **Re** $\ll 1$: viscous diffusion of momentum acts quickly compared with advection. The fluid “knows” the body is there before the fluid element reaches it — smooth, reversible Stokes flow.
- **Re** $\gg 1$: the fluid element sweeps past the body before momentum diffusion can communicate the body’s presence to nearby fluid. Advection dominates; flow separates and turbulence develops.

6.9 Summary of the Unified Transport Picture

The Physical Picture: All three classical transport phenomena — diffusion, heat conduction, and viscosity — share a common mechanism: microscopic carriers (molecules, electrons, phonons) performing random walks and transporting conserved quantities across gradients. Dimensional analysis immediately gives the transport coefficients; the Prandtl number then reveals whether one mechanism dominates or whether different carriers are responsible for different quantities.

- Particle diffusion: $J = -D\nabla n$, $D \sim \ell v_{\text{th}}$, $\partial_t n = D\nabla^2 n$.
- Heat conduction: $J_Q = -\kappa\nabla T$, $\kappa = D c_V$, $\partial_t T = D_{\text{th}}\nabla^2 T$.
- Viscous flow: $\sigma_{xy} = -\eta\partial_y v_x$, $\nu = \eta/\rho \sim \ell v_{\text{th}}$, full Navier-Stokes equation (Eq. (66)).
- Diffusion timescale: $\tau \sim L^2/D$ (quadratic in L — very slow at large scales, replaced by turbulence in real systems).
- Prandtl number: $\text{Pr} = \nu/D_{\text{th}}$; air ≈ 0.7 , mercury ≈ 0.025 , polymers $\gg 1$.
- Reynolds number: $\text{Re} = VR/\nu = \tau_{\text{diff}}/\tau_{\text{cross}}$; low Re gives Stokes flow $F = 6\pi\eta VR$, high Re gives turbulent drag $F \sim \rho V^2 R^2$.

7 Lecture 7: Convection, Atmospheric Stability, and Dimensionless Control

Lecture date: 28/05/2026

The previous lecture built a unified picture of transport by microscopic random walks: diffusion of particles, diffusion of heat, and diffusion of momentum. This lecture turns to a different kind of heat transport: *convection*. Instead of heat being carried by microscopic steps, a whole fluid element moves, expands or contracts, and mixes with its surroundings. Gravity is the essential new ingredient, because it lets density differences turn into buoyant acceleration.

7.1 Three Ways to Transport Heat

Physical Motivation

Before asking when convection starts, it is useful to separate it from the other two heat-transfer mechanisms. A warm object in a room can exchange heat with its surroundings by:

- **conduction:** molecular collisions transfer energy through matter, described by Fourier's law;
- **radiation:** photons carry energy, even through vacuum;
- **convection:** the material itself moves, so hot and cold parcels are physically exchanged.

Convection is therefore not just “better conduction”. It changes the configuration of the fluid.

In the Earth's atmosphere, sunlight mostly reaches the surface directly because the atmosphere is relatively transparent at visible wavelengths. The warmed surface then emits infrared radiation, to which the atmosphere is much more opaque. The lower atmosphere is heated from below, and the question becomes: will the heat escape by slow conduction/radiation, or will the air itself begin to rise and overturn?

Physical Concept 7.1: Convection

Convection is heat transport by the bulk motion of a fluid. In a gravitational field it is driven by buoyancy: a parcel whose density is lower than the density of its surroundings accelerates upward, while a denser parcel accelerates downward.

7.2 Hydrostatic Balance and an Isothermal Atmosphere

Mathematical Setup

Let z denote height, $P(z)$ pressure, $\rho(z)$ mass density, $T(z)$ temperature, g the gravitational acceleration, and μ the mass per gas particle. A static atmosphere must balance gravity against the pressure gradient. Per unit mass,

$$\frac{1}{\rho} \frac{dP}{dz} = -g. \quad (77)$$

Significance: Pressure must decrease upward because lower layers support the weight of all the gas above them.

For an ideal gas,

$$P = nk_B T = \frac{\rho}{\mu} k_B T, \quad (78)$$

where $n = \rho/\mu$ is the particle number density. Equations (77) and (78) are not yet enough to determine the atmosphere, because both $\rho(z)$ and $T(z)$ are unknown. One possible closure is an isothermal atmosphere, $T(z) = T_0$.

Substituting $P = \rho k_B T_0 / \mu$ into hydrostatic balance gives

$$\frac{1}{\rho} \frac{d}{dz} \left(\frac{\rho k_B T_0}{\mu} \right) = -g, \quad (79)$$

$$\frac{k_B T_0}{\mu} \frac{1}{\rho} \frac{d\rho}{dz} = -g, \quad (80)$$

$$\frac{d \ln \rho}{dz} = -\frac{\mu g}{k_B T_0}. \quad (81)$$

Integrating from $z = 0$ to z ,

$$\boxed{\rho(z) = \rho_0 e^{-z/H}, \quad H = \frac{k_B T_0}{\mu g}.} \quad (82)$$

Significance: H is the atmospheric scale height: it is the height over which the density falls by a factor of e .

For Earth, taking $T_0 \simeq 300$ K, $\mu \simeq 29m_p$, and $g \simeq 10 \text{ m s}^{-2}$ gives

$$H \sim 8 \text{ km}. \quad (83)$$

This does not mean that the atmosphere ends after 8 km; it means that the density has dropped by one e -fold.

7.3 Adiabatic Atmospheres and the Lapse Rate

Conceptual Background

Convection is tested by displacing a fluid element. Pressure equilibrates quickly, on roughly a sound-crossing time, while heat exchange by conduction or radiation can be much slower. Therefore a rapidly displaced parcel is often approximately *adiabatic*: it changes pressure, density, and temperature without exchanging heat with its surroundings.

For an ideal gas undergoing adiabatic changes,

$$P = K \rho^\gamma, \quad (84)$$

where K is constant along the parcel trajectory and $\gamma = c_p/c_V$ is the adiabatic index. Combining this with $P = \rho k_B T / \mu$ gives

$$T \propto \rho^{\gamma-1}, \quad P \propto T^{\gamma/(\gamma-1)}. \quad (85)$$

Differentiate the ideal-gas pressure relation along an adiabatic profile:

$$dP = \gamma \frac{P}{\rho} d\rho, \quad (86)$$

$$dT = \frac{\mu}{k_B} \left(\frac{dP}{\rho} - \frac{P}{\rho^2} d\rho \right). \quad (87)$$

Using $d\rho = (\rho/\gamma P) dP$ in Eq. (87),

$$dT = \frac{\mu}{k_B} \left(\frac{1}{\rho} - \frac{1}{\gamma \rho} \right) dP = \frac{\mu}{k_B} \frac{\gamma - 1}{\gamma} \frac{dP}{\rho}. \quad (88)$$

Hydrostatic balance gives $dP/dz = -\rho g$, hence

$$\boxed{\left(\frac{dT}{dz} \right)_{\text{ad}} = -\frac{\gamma - 1}{\gamma} \frac{\mu g}{k_B} = -\frac{g}{c_p}.} \quad (89)$$

Significance: A dry parcel cools as it rises because it expands and does work on its surroundings. The atmospheric temperature gradient cannot be arbitrarily steep without triggering convection.

For dry air, this gives roughly

$$\left| \frac{dT}{dz} \right|_{\text{ad}} \sim 10 \text{ K km}^{-1} \simeq 1^\circ \text{C}/100 \text{ m}. \quad (90)$$

Note:

Moist air has a smaller effective adiabatic lapse rate. As a rising parcel expands and cools, water vapor can condense and release latent heat. That heat partly offsets the cooling, so saturated air can become convectively unstable for a weaker temperature gradient than dry air.

7.4 Convective Stability as an Entropy Criterion

Physical Motivation

It is tempting to say that an atmosphere is stable because dense air is below light air. That is not sufficient. A displaced parcel does not keep its original density: it expands or contracts to match the ambient pressure. The right comparison is made after the parcel has adjusted its pressure but before it has exchanged heat. In other words, the parcel keeps its entropy.

For an ideal gas, the entropy per unit mass can be written, up to an additive constant, as

$$s = c_V \ln \left(\frac{P}{\rho^\gamma} \right) = c_p \ln T - \frac{k_B}{\mu} \ln P + \text{const.} \quad (91)$$

An adiabatic displacement keeps s fixed. Suppose a parcel from height z is lifted slightly to $z + dz$. It rapidly adjusts to the new pressure $P(z + dz)$, but retains $s(z)$. Then:

- if the environmental entropy increases upward, $ds/dz > 0$, the lifted parcel has lower entropy than its new surroundings; at the same pressure it is denser, so it sinks back;
- if the environmental entropy decreases upward, $ds/dz < 0$, the lifted parcel has higher entropy than its new surroundings; at the same pressure it is lighter, so it keeps rising.

Thus

$$\frac{ds}{dz} > 0 \implies \text{stable}, \quad \frac{ds}{dz} < 0 \implies \text{convectively unstable.} \quad (92)$$

Significance: Convection begins when the one-dimensional stratified solution is unstable to vertical displacements; the system then develops overturning motions instead of staying purely radial or vertical.

Using Eq. (89), the same criterion can be phrased as a temperature-gradient condition. Define the environmental lapse rate

$$\Gamma \equiv -\frac{dT}{dz}, \quad \Gamma_{\text{ad}} \equiv \frac{g}{c_p}. \quad (93)$$

Then

$$\Gamma < \Gamma_{\text{ad}} \implies \text{stable}, \quad \Gamma > \Gamma_{\text{ad}} \implies \text{convectively unstable.} \quad (94)$$

Significance: A temperature that falls with height faster than the adiabatic rate makes rising parcels remain warmer and lighter than their surroundings.

In ordinary dry air, the condition says that height alone should not produce a temperature difference much larger than about 1°C per 100 m. If the gradient becomes steeper, convection mixes the air and drives the profile back toward marginal stability.

7.5 Why Efficient Convection Drives Marginal Stability

Physical Motivation

Convection often looks like a violent instability, but in many systems it self-regulates. If the entropy gradient is strongly unstable, buoyant motions become fast and carry heat very efficiently. That heat transport reduces the entropy contrast that caused the motion. The final state is usually close to the boundary between stable and unstable.

This explains why convective regions in stars, atmospheres, and planetary interiors are often nearly adiabatic. They are not exactly adiabatic, because a small superadiabatic excess is needed to carry the heat flux, but the excess can be tiny when convection is efficient.

Examples discussed in the lecture:

- **Earth's atmosphere:** surface heating can make the lower atmosphere unstable; overturning then limits the lapse rate.
- **Earth's mantle:** the mantle can convect and carry heat from the hot interior, while the rigid crust conducts heat across the final boundary layer.
- **the Sun:** convection in the outer envelope transports heat by bulk motion; if the whole Sun were strongly superadiabatic, it would mix on a sound-crossing timescale of only hours, so the actual convective stratification is close to marginal.

7.6 Buoyancy and a Convective Velocity Scale

Mathematical Setup

Let α be the thermal expansion coefficient,

$$\alpha \equiv -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_P. \quad (95)$$

For a small temperature excess ΔT at fixed pressure,

$$\frac{\Delta \rho}{\rho} \simeq -\alpha \Delta T. \quad (96)$$

For an ideal gas, $\alpha = 1/T$.

Consider a parcel with density ρ_1 embedded in surroundings of density ρ_2 . The gravitational force on a volume V is $\rho_1 V g$ downward, while the buoyant force is $\rho_2 V g$ upward. The net acceleration is

$$a = g \frac{\rho_2 - \rho_1}{\rho_1} \sim g \frac{\Delta \rho}{\rho}. \quad (97)$$

Using Eq. (96),

$$\boxed{a_{\text{buoy}} \sim g \alpha \Delta T_{\text{ex}}}. \quad (98)$$

Significance: Only the temperature excess beyond the adiabatic background contributes to buoyant acceleration; the ordinary adiabatic cooling of a lifted parcel is already part of the neutral reference state.

If the unstable layer has height H , the buoyancy time and the corresponding free convective velocity scale are

$$\tau_{\text{buoy}} \sim \sqrt{\frac{H}{g\alpha\Delta T_{\text{ex}}}}, \quad (99)$$

$$U_{\text{conv}} \sim \frac{H}{\tau_{\text{buoy}}} \sim \sqrt{g\alpha\Delta T_{\text{ex}}H}. \quad (100)$$

Significance: Larger gravity, stronger thermal expansion, larger superadiabatic temperature contrast, and taller layers all make convection faster.

7.7 Dimensionless Numbers for Convection

Motivation

The final part of the lecture returns to dimensional analysis. A convective layer is controlled by the layer height H , the superadiabatic temperature contrast ΔT_{ex} , gravity g , thermal expansion α , kinematic viscosity ν , and thermal diffusivity χ . Both ν and χ have units of m^2s^{-1} , but they diffuse different quantities: momentum and heat.

Physical Concept 7.2: Thermal Diffusivity

The thermal diffusivity χ is the diffusion coefficient for temperature:

$$\frac{\partial T}{\partial t} = \chi \nabla^2 T.$$

It should be distinguished from particle diffusion and from kinematic viscosity, even though all three have dimensions of length squared over time.

The first dimensionless number is the Prandtl number:

$$\text{Pr} = \frac{\nu}{\chi}. \quad (101)$$

Significance: Pr compares how fast momentum diffuses to how fast heat diffuses.

The second important comparison is between buoyant driving and diffusive damping. The heat-diffusion time across the layer is

$$\tau_{\chi} \sim \frac{H^2}{\chi}, \quad (102)$$

and the viscous momentum-diffusion time is

$$\tau_{\nu} \sim \frac{H^2}{\nu}. \quad (103)$$

Combining these with the buoyancy time (99) gives the Rayleigh number:

$$\text{Ra} = \frac{g\alpha\Delta T_{\text{ex}}H^3}{\nu\chi}. \quad (104)$$

Significance: Ra measures buoyant driving against the two diffusive processes that suppress convection: viscosity damps velocity, and thermal diffusion erases the temperature anomaly that produces buoyancy.

Equivalently,

$$\text{Ra} = \frac{\tau_\nu \tau_\chi}{\tau_{\text{buoy}}^2}. \quad (105)$$

Large Ra means buoyancy acts before diffusion can smooth the temperature and momentum perturbations. Small Ra means diffusion wins, and the layer remains conductive. In the standard Rayleigh-Benard problem with fixed-temperature plates, convection starts at $\text{Ra} \simeq 1708$; for order-of-magnitude modeling the important point is the scaling $\text{Ra} \gtrsim 1$ versus $\text{Ra} \ll 1$.

Relation to Reynolds number

Using the convective velocity estimate (100), the Reynolds number of the convective motion is

$$\text{Re}_{\text{conv}} = \frac{U_{\text{conv}} H}{\nu} \sim \frac{H}{\nu} \sqrt{g \alpha \Delta T_{\text{ex}} H}. \quad (106)$$

Squaring and using Eq. (104),

$$\boxed{\text{Re}_{\text{conv}}^2 \sim \frac{\text{Ra}}{\text{Pr}}}. \quad (107)$$

Significance: The speed of convective stirring is not an externally given input like the speed of a stirred cup; it is part of the solution, set by buoyancy and limited by diffusion.

7.8 Summary of the Lecture

- Hydrostatic balance fixes $dP/dz = -\rho g$, but does not by itself determine $T(z)$ and $\rho(z)$.
- An isothermal ideal-gas atmosphere has $\rho = \rho_0 e^{-z/H}$ with $H = k_B T / (\mu g)$.
- A rapidly displaced parcel evolves adiabatically, so the neutral temperature gradient is $dT/dz = -g/c_p$.
- The stability criterion is entropy stratification: $ds/dz > 0$ is stable, while $ds/dz < 0$ is convectively unstable.
- The practical atmospheric form is $\Gamma < \Gamma_{\text{ad}}$ stable and $\Gamma > \Gamma_{\text{ad}}$ unstable, with $\Gamma_{\text{ad}} \simeq 10 \text{ K km}^{-1}$ for dry air.
- Buoyant acceleration is $a_{\text{buoy}} \sim g \alpha \Delta T_{\text{ex}}$, where ΔT_{ex} is the superadiabatic temperature excess.
- The main dimensionless control parameters are $\text{Pr} = \nu/\chi$ and $\text{Ra} = g \alpha \Delta T_{\text{ex}} H^3 / (\nu \chi)$.

8 Homework 1: Estimates

8.1 Piano Tuners in Chicago

Question: How many piano tuners are there, in your estimation, in the city of Chicago?

Solution: We decompose the problem into several factors: the population of Chicago, the fraction of households with a piano, and the frequency of tunings, compared to the working capacity of a single tuner.

1. **Pianos in Chicago:** The population of Chicago is roughly 3×10^6 . Assuming an average household size of 3 people, there are about 10^6 households. If we estimate that roughly 1 in 10 households owns a piano, there are $N_{\text{pianos}} \approx 10^5$ pianos in the city.
2. **Tunings required per year:** A piano typically requires tuning once a year. Thus, there are about 10^5 tunings required per year.
3. **Tuner's capacity:** A typical tuner might tune 2 pianos a day. Assuming they work 5 days a week for 50 weeks a year (250 working days), one tuner can perform roughly $2 \times 250 = 500$ tunings per year.

Therefore, the number of piano tuners required is:

$$N_{\text{tuners}} \approx \frac{10^5 \text{ tunings/year}}{500 \text{ tunings/year/tuner}} = 200 \text{ tuners.}$$

8.2 Grains of Sand in Israel's Beaches

Question: How many grains of sand are there on the beaches of Israel? How many houses would this amount suffice to build?

Solution: First, we estimate the total volume of sand on the beaches.

- The coastline of Israel is approximately $200 \text{ km} = 2 \times 10^5 \text{ m}$ long.
- The average width of a sandy beach is roughly 50 m .
- The depth of the sand layer is perhaps 10 m .

The total volume of beach sand is:

$$V_{\text{sand}} \approx (2 \times 10^5) \times 50 \times 10 = 10^8 \text{ m}^3.$$

Next, we estimate the volume of a single grain of sand. Assuming a grain is a small cube of side length $1 \text{ mm} = 10^{-3} \text{ m}$, its volume is:

$$v_{\text{grain}} \approx (10^{-3} \text{ m})^3 = 10^{-9} \text{ m}^3.$$

The total number of grains of sand is therefore:

$$N_{\text{grains}} = \frac{V_{\text{sand}}}{v_{\text{grain}}} \approx \frac{10^8}{10^{-9}} = 10^{17} \text{ grains.}$$

For the second part of the question, we estimate the volume of sand needed to build a standard house (primarily for concrete). A typical house might require about 50 m^3 of sand. The number of houses that could be built is:

$$N_{\text{houses}} = \frac{V_{\text{sand}}}{50 \text{ m}^3/\text{house}} = \frac{10^8}{50} = 2 \times 10^6 \text{ houses.}$$

8.3 Mass of the Earth's Atmosphere

Question: Estimate the total mass of the Earth's atmosphere.

Solution: Atmospheric pressure at sea level is caused by the weight of the air column directly above. The pressure is $P \approx 10^5$ Pa (N/m²). By definition, pressure is force per unit area: $P = \frac{F}{A} = \frac{M_{\text{atm}}g}{4\pi R_{\oplus}^2}$. Solving for the mass of the atmosphere M_{atm} :

$$M_{\text{atm}} = \frac{P \cdot 4\pi R_{\oplus}^2}{g}.$$

Using the Earth's radius $R_{\oplus} \approx 6.4 \times 10^6$ m and $g \approx 10$ m/s²:

$$A_{\oplus} = 4\pi(6.4 \times 10^6)^2 \approx 12.5 \times 40 \times 10^{12} \approx 5 \times 10^{14} \text{ m}^2.$$

$$M_{\text{atm}} \approx \frac{10^5 \times 5 \times 10^{14}}{10} = 5 \times 10^{18} \text{ kg}.$$

8.4 Value of Your Time

Question: How much is your time worth on average? (How much will you earn in a lifetime? How long will you live?)

Solution: We estimate the value of an average hour of an Israeli's life.

- **Lifetime Earnings:** The average monthly salary in Israel is roughly 12,000 NIS, which is about 150,000 NIS/year. Over a 40-year working career, the total lifetime earnings are $\sim 6 \times 10^6$ NIS.
- **Total Lifespan:** An average life expectancy of 80 years corresponds to $80 \times 365 \times 24 \approx 7 \times 10^5$ hours.

The average financial value of one hour of life is:

$$\text{Value of living hour} = \frac{6 \times 10^6 \text{ NIS}}{7 \times 10^5 \text{ hours}} \approx 8.5 \text{ NIS/hour}.$$

If we restrict the calculation to only *working hours* (approx. 2000 hours/year for 40 years = 80,000 hours), the value of a working hour is $\frac{6 \times 10^6}{8 \times 10^4} = 75$ NIS/hour, consistent with the average wage.

8.5 Car Accidents and Life Expectancy

Question: By how much do car accidents shorten life expectancy? If the chance of dying from any cause before age 30 is roughly twice the risk from car accidents alone, how much should a \$1 million life insurance policy cost you?

Solution: Part 1: Shortened life expectancy. There are approximately 300 fatalities from car accidents per year in Israel, out of a population of $\sim 9 \times 10^6$. The annual risk of dying in a car accident is:

$$p_{\text{annual}} \approx \frac{300}{9 \times 10^6} \approx 3.3 \times 10^{-5}.$$

Over an 80-year lifetime, the cumulative risk of dying in a car crash is $80 \times 3.3 \times 10^{-5} \approx 2.6 \times 10^{-3}$ (or about 0.26%). If an average car crash victim loses about 40 years of life, the expected loss of life expectancy across the population is:

$$\Delta t \approx 40 \text{ years} \times 2.6 \times 10^{-3} \approx 0.1 \text{ years} \approx 1 \text{ month}.$$

Part 2: Life insurance cost. The risk of dying from a car accident before age 30 is approximately $30 \times 3.3 \times 10^{-5} \approx 10^{-3}$. We are given that the total risk of dying from *any* cause before age 30 is roughly twice this, so:

$$P(\text{die before 30}) \approx 2 \times 10^{-3} = 0.2\%.$$

The actuarially fair premium for a \$1,000,000 life insurance policy covering this period should be equal to the expected payout:

$$\text{Cost} = \$1,000,000 \times 2 \times 10^{-3} = \$2,000.$$

8.6 Hairs on Your Head

Question: How many hairs are on your head?

Solution: The scalp can be modeled as roughly a hemisphere of radius $r \approx 10$ cm. Its surface area is:

$$A \approx 2\pi r^2 \approx 2 \times 3 \times 100 \text{ cm}^2 = 600 \text{ cm}^2 = 60,000 \text{ mm}^2.$$

By looking closely at the scalp, we can estimate a hair density of roughly 1 hair per square millimeter (~ 100 per cm^2). The total number of hairs is:

$$N_{\text{hair}} = A \times \text{density} = 60,000 \text{ mm}^2 \times 1 \text{ hair/mm}^2 = 60,000 \text{ hairs}.$$

(This aligns reasonably well with biological averages, which range from 90,000 to 150,000).

8.7 Israel's Fuel Consumption and Inflation

Question: Estimate the fuel consumption of the State of Israel. If the price of oil rises from \$80 to \$115 a barrel, what will be the expected inflation? (What fraction of the price of fuel is actually the fuel?)

Solution: Fuel Consumption: Assume there are $\sim 3 \times 10^6$ cars in Israel. An average car travels $\sim 15,000$ km/year and has a fuel efficiency of ~ 10 km/L. The total annual gasoline consumption is:

$$V_{\text{fuel}} \approx 3 \times 10^6 \text{ cars} \times \frac{15,000 \text{ km/year}}{10 \text{ km/L}} = 4.5 \times 10^9 \text{ Liters/year}.$$

Using the conversion 1 barrel ≈ 160 L, this is roughly $\sim 3 \times 10^7$ barrels/year.

Inflation Estimate: If oil rises from \$80 to \$115 per barrel, the price increase is $\Delta P = \$35/\text{barrel}$. The additional macroeconomic cost to the country is:

$$\Delta \text{Cost} = 3 \times 10^7 \text{ barrels} \times \$35 \approx \$10^9.$$

The GDP of Israel is roughly $\$500 \times 10^9$. The expected direct contribution to inflation is the additional cost divided by the GDP:

$$\text{Inflation contribution} \approx \frac{10^9}{500 \times 10^9} = 0.002 = 0.2\%.$$

Fraction of the price that is raw fuel: At \$80 per barrel (160 L), the cost of raw unrefined oil is \$0.50 per Liter, or roughly 1.8 NIS/L. Given that the price of gasoline at the pump in Israel is around 7.5 NIS/L, the actual fuel accounts for only:

$$\frac{1.8 \text{ NIS}}{7.5 \text{ NIS}} \approx 25\%$$

of the price. The remaining 75% consists of taxes, refining, and distribution costs.

9 Homework 2: Scaling

9.1 Maximum Height of Mountains

Question: (a) Estimate the maximum possible height for a mountain on Earth. How does the result change if most of the mountain is underwater? (b) Write a scaling law for the maximum mountain height on an arbitrary planet. (c) For what scale of planetary radius will mountains be roughly the size of the planet itself?

Solution: (a) Mountain on Earth: The maximum height of a mountain is limited by the yield strength of the rock at its base. If the pressure exceeds the critical yield stress $P_{\max}$, the rock will undergo plastic deformation, causing the mountain to sink. The pressure at the base is $P = \rho_{\text{rock}} g H$. Equating this to $P_{\max}$:

$$H_{\max} \approx \frac{P_{\max}}{\rho_{\text{rock}} g}.$$

For typical rock, $P_{\max} \sim 10^8$ Pa and $\rho_{\text{rock}} \approx 3000$ kg/m³. With $g \approx 10$ m/s²:

$$H_{\max} \approx \frac{10^8}{3000 \times 10} \approx 3000 \text{ m} = 3 \text{ km}.$$

(While mountains like Everest are taller, they are supported by crustal roots floating on the mantle—*isostasy*—but the basic scale is correct). If the mountain is underwater, buoyant forces reduce the effective downward force. The effective density becomes $\Delta\rho = \rho_{\text{rock}} - \rho_{\text{water}} \approx 3000 - 1000 = 2000$ kg/m³. Thus, underwater mountains can be taller by a factor of $\frac{3000}{2000} = 1.5$, giving $H_{\max} \approx 4.5$ km.

(b) Scaling Law for Arbitrary Planet: The surface gravity of a planet of radius R and average density $\bar{\rho}_p$ is $g_p \propto \bar{\rho}_p R$. Substituting this into the mountain height equation yields the scaling relation:

$$H_{\max} \propto \frac{P_{\max}}{\rho_{\text{rock}} \bar{\rho}_p R}.$$

(c) Mountains the Size of the Planet: We set $H_{\max} \sim R$:

$$R \sim \frac{P_{\max}}{\rho_{\text{rock}} \bar{\rho}_p G R} \implies R \sim \sqrt{\frac{P_{\max}}{G \rho_{\text{rock}} \bar{\rho}_p}}.$$

Assuming $\bar{\rho}_p \approx \rho_{\text{rock}} \approx 3000$ kg/m³:

$$R \sim \sqrt{\frac{10^8}{6.67 \times 10^{-11} \times 3000^2}} \approx \sqrt{1.6 \times 10^{11}} \approx 400 \text{ km}.$$

This is a very logical result: bodies smaller than ~ 400 km (like many asteroids) have insufficient gravity to overcome the rock's rigidity, allowing them to remain highly irregular ("mountains" the size of the asteroid itself). Larger bodies will crush their own irregularities and pull themselves into spheres.

9.2 Air Drag and Scaling

Question: Air drag is given by $F_d = \frac{1}{2} C_d \rho_{\text{air}} v^2 A$. (a) Find a scaling relation for terminal velocity depending on body size and density, relative to a reference body. (b) Find an

expression for fuel consumption depending on velocity. How much does it increase if the speed limit is raised by 10 km/h? (c) Estimate the relative air density at 10 km given plane and car fuel prices.

Solution: (a) Terminal Velocity Scaling: At terminal velocity, the drag force balances gravity: $F_d = Mg$. Using $M \propto \rho_{\text{body}} a^3$ and $A \propto a^2$ (where a is the characteristic size of the body):

$$\rho_{\text{body}} a^3 g \propto C_d \rho_{\text{air}} v^2 a^2 \implies v^2 \propto \frac{\rho_{\text{body}} g a}{C_d \rho_{\text{air}}}.$$

Relative to a reference body (a_0, ρ_0, v_0) falling in the same air, and with comparable shape,

$$v = v_0 \sqrt{\frac{\rho_{\text{body}}}{\rho_0} \frac{a}{a_0}}.$$

More generally one should also include the ratio $\sqrt{(C_{d,0} \rho_{\text{air},0}) / (C_d \rho_{\text{air}})}$.

(b) Fuel Consumption vs. Velocity: Fuel consumption is energy expended per unit distance, $\frac{E}{d}$. Since work is $E = F \cdot d$, the fuel consumption is directly proportional to the drag force:

$$\text{Consumption} \propto F_d \propto v^2.$$

If the speed limit on a highway is increased from 90 km/h to 100 km/h, the velocity increases by a factor of $\frac{100}{90} \approx 1.11$. The fuel consumption will scale as $1.11^2 \approx 1.23$, an increase of roughly 23%.

(c) Air Density at 10 km: The work done per passenger over a trip of length L is

$$W \propto F_d L \propto \rho_{\text{air}} v^2 A_{\text{per passenger}} L.$$

For a Jumbo jet at 10 km altitude, the fuel cost per passenger from New York to Tel Aviv is about \$150 for $L \sim 10^4$ km. For a car at sea level, 5 L/100 km over the same distance gives 500 L. If car fuel costs about half the airplane-fuel price, the car's fuel energy cost in "airplane-fuel dollars" is equivalent to about 250 liters-worth, so

$$\frac{W_{\text{plane}}}{W_{\text{car}}} \sim \frac{150}{250} \sim 0.6.$$

Now compare drag work:

$$\frac{W_{\text{plane}}}{W_{\text{car}}} \sim \frac{\rho_{10}}{\rho_0} \left(\frac{v_{\text{plane}}}{v_{\text{car}}} \right)^2 \frac{A_{\text{plane/pass}}}{A_{\text{car/pass}}}.$$

Use $v_{\text{plane}} \sim 900$ km/h, $v_{\text{car}} \sim 100$ km/h, so the velocity factor is ~ 80 . A Jumbo cross-section of order 50 m^2 divided among a few hundred passengers gives $A_{\text{plane/pass}} \sim 0.1\text{--}0.2 \text{ m}^2$, while a car carries one to several people with $A_{\text{car/pass}} \sim 0.5\text{--}2 \text{ m}^2$. Therefore $A_{\text{car/pass}}/A_{\text{plane/pass}} \sim 5\text{--}20$, and

$$\frac{\rho_{10}}{\rho_0} \sim 0.6 \left(\frac{100}{900} \right)^2 \frac{A_{\text{car/pass}}}{A_{\text{plane/pass}}} \sim 0.05\text{--}0.2.$$

Given the very crude economics and geometry, this is correctly of order a tenth to a few tenths of sea-level density, close to the real value $\rho_{10}/\rho_0 \sim 0.25\text{--}0.35$.

9.3 Bomb Damage Scaling

Question: A bomb causes damage at distance R . At what distance will it cause damage larger by a factor of k ?

Solution: Assume the relevant damage measure is set by blast energy density or overpressure scale, and that at radius r the explosion energy is spread over a volume of order r^3 :

$$u(r) \sim \frac{E}{r^3}.$$

If the same bomb causes damage D at distance R , then at a different distance r the damage proxy scales like

$$\frac{D(r)}{D(R)} \sim \frac{u(r)}{u(R)} = \left(\frac{R}{r}\right)^3.$$

For damage larger by a factor k ,

$$k = \left(\frac{R}{r}\right)^3 \implies \boxed{r = R k^{-1/3}}.$$

Equivalently, to cause the same damage at a distance larger by a factor k , the explosion energy must be increased by a factor k^3 .

9.4 Magnetic Field Scaling of a Collapsing Star

Question: A star has a dipole magnetic field B . The star contracts from $R \rightarrow \alpha R$. What is the new field B' under three different conservation scenarios?

Solution: (a) Total magnetic flux is conserved: The magnetic flux is $\Phi \propto BR^2$. If Φ is constant, then $B'(\alpha R)^2 = BR^2$.

$$B' = \alpha^{-2}B.$$

(b) Total magnetic energy is conserved: The energy in the magnetic field is $E_B \propto B^2V \propto B^2R^3$. If E_B is constant, then $(B')^2(\alpha R)^3 = B^2R^3$.

$$(B')^2 = \alpha^{-3}B^2 \implies B' = \alpha^{-3/2}B.$$

(c) Ratio of magnetic to gravitational energy is conserved: The gravitational binding energy of the star is $E_G \propto \frac{GM^2}{R}$. Assuming the mass M of the star is conserved during the collapse, $E_G \propto R^{-1}$. We are given that $\frac{E_B}{E_G} = \text{const}$.

$$\frac{B^2R^3}{R^{-1}} = \text{const} \implies B^2R^4 = \text{const}.$$

Therefore, $(B')^2(\alpha R)^4 = B^2R^4$.

$$B' = \alpha^{-2}B.$$

10 Homework 3: Dimensional Analysis and Dimensionless Numbers

10.1 Pendula and a Hanging Rope

Question: A pendulum has mass m and length ℓ . By dimensional reasoning, what is the most general expression for the oscillation frequency? What changes for a physical pendulum with moment of inertia I ? How long does a disturbance take to travel from one end to the other of a rope of mass m and length ℓ hanging from the ceiling?

Solution: For a simple pendulum the relevant dimensional quantities are g and ℓ ; the mass cannot affect the answer because it introduces no restoring scale independent of gravity. Thus

$$\omega = C\sqrt{\frac{g}{\ell}}, \quad T = \frac{2\pi}{\omega} \sim \sqrt{\frac{\ell}{g}},$$

with $C = 1$ for small oscillations of an ideal simple pendulum.

For a physical pendulum, the restoring torque is $\tau \sim Mgd\theta$, where d is the distance from the pivot to the center of mass, and the angular inertia is I . Hence

$$I\ddot{\theta} + Mgd\theta \simeq 0, \quad \boxed{\omega \sim \sqrt{\frac{Mgd}{I}}}.$$

For a hanging rope with linear density $\lambda = m/\ell$, the tension a distance x above the free lower end is the weight below that point:

$$T(x) = \lambda gx.$$

The local transverse wave speed is therefore

$$c(x) = \sqrt{\frac{T(x)}{\lambda}} = \sqrt{gx}.$$

The travel time from bottom to top is

$$t = \int_0^\ell \frac{dx}{c(x)} = \int_0^\ell \frac{dx}{\sqrt{gx}} = 2\sqrt{\frac{\ell}{g}}.$$

10.2 Oscillation Through the Earth

Question: A tunnel is drilled through Earth. Assuming uniform density ρ , what is the oscillation period through the center? What if the motion starts at radius r ?

Solution: Inside a uniform sphere only the mass interior to radius r contributes to the gravitational force:

$$M(r) = \frac{4\pi}{3}\rho r^3, \quad F = -\frac{GmM(r)}{r^2} = -\frac{4\pi G\rho}{3}mr.$$

This is exactly a harmonic oscillator,

$$\ddot{r} = -\omega^2 r, \quad \omega^2 = \frac{4\pi G\rho}{3}.$$

Therefore

$$T = 2\pi \sqrt{\frac{3}{4\pi G\rho}}.$$

The period is independent of amplitude, so if the motion starts from any radius r inside the Earth, the amplitude changes but the period does not. For Earth's mean density this gives about 80–90 minutes.

10.3 Magnetic Pressure

Question: Given a magnetic field B , what pressure or tension can be built from it? Explain the dependence on SI versus CGS units.

Solution: The magnetic pressure is the magnetic energy density. In SI units,

$$P_B = u_B = \frac{B^2}{2\mu_0}.$$

In Gaussian-CGS units,

$$P_B = u_B = \frac{B^2}{8\pi}.$$

The field also produces magnetic tension along field lines, of the same order as the magnetic pressure. More precisely the Maxwell stress contains a pressure term $B^2/(2\mu_0)$ in SI and a line-tension term B^2/μ_0 in SI (or $B^2/4\pi$ in CGS), with geometry deciding which component is relevant.

10.4 Radiation from Dipoles and Quadrupoles

Question: A magnetic dipole moment m varies periodically with frequency ω . Use dimensional analysis to find the radiated power. What about an electric quadrupole and a gravitational quadrupole?

Solution: As in the lecture derivation of Larmor radiation, we work up to factors of order unity. For an oscillating electromagnetic dipole, the radiated power is quadratic in the second time derivative of the dipole moment:

$$P_{\text{mag dipole}} \sim \frac{m^2 \omega^4}{c^3}.$$

An electric quadrupole has one extra power of length compared with a dipole. Radiation begins from the third time derivative of the quadrupole moment, so

$$P_{\text{electric quad}} \sim \frac{Q_e^2 \omega^6}{c^5}.$$

For gravity, monopole radiation is forbidden by mass conservation and dipole radiation by momentum conservation, so the leading term is quadrupolar:

$$P_{\text{grav quad}} \sim \frac{G}{c^5} Q_g^2 \omega^6,$$

where $Q_g \sim ML^2$ is the mass quadrupole moment.

10.5 Compton Wavelength and Cross Section

Question: What is the electron Compton wavelength? What cross section can be built from it, and what is the analogous proton result?

Solution: Set the photon energy $\hbar c/\lambda$ equal to the electron rest energy $m_e c^2$:

$$\lambda_C = \frac{\hbar}{m_e c} \simeq 3.9 \times 10^{-13} \text{ m}.$$

The only geometric cross section that can be built from this length is

$$\sigma_C \sim \lambda_C^2 \simeq 1.5 \times 10^{-25} \text{ m}^2.$$

This is the natural Compton-scale area. In detailed low-energy electromagnetic scattering one obtains the Thomson cross section, which is smaller by powers of the fine-structure constant because the coupling strength matters in addition to dimensional analysis.

For a proton,

$$\lambda_{C,p} = \frac{\hbar}{m_p c} = \lambda_{C,e} \frac{m_e}{m_p} \simeq \frac{3.9 \times 10^{-13}}{1836} \text{ m} \simeq 2.1 \times 10^{-16} \text{ m},$$

and therefore

$$\sigma_{C,p} \sim \sigma_{C,e} \left(\frac{m_e}{m_p} \right)^2 \sim 4 \times 10^{-32} \text{ m}^2.$$

10.6 Planck Energy and Planck Mass

Question: Estimate the energy at which quantum and gravitational effects are equally important. Estimate the corresponding mass.

Solution: Quantum mechanics gives a Compton length $\lambda_C = \hbar/(mc)$. Gravity gives a Schwarzschild length $r_s \sim Gm/c^2$. The two become comparable when

$$\frac{\hbar}{mc} \sim \frac{Gm}{c^2} \implies m_P \sim \sqrt{\frac{\hbar c}{G}}.$$

Numerically,

$$m_P \simeq 2.2 \times 10^{-8} \text{ kg}, \quad E_P = m_P c^2 \simeq 2 \times 10^9 \text{ J} \simeq 1.2 \times 10^{19} \text{ GeV}.$$

This is the Planck scale: the energy scale at which a particle's quantum wavelength is comparable to its gravitational radius.

10.7 Central Pressure of Earth and Sun

Question: Estimate the pressure at the center of Earth and the Sun using dimensional analysis. Since the Sun is a gas, estimate its central temperature.

Solution: Hydrostatic balance gives $dP/dr \sim \rho g$. With $g \sim GM/R^2$ and $\rho \sim M/R^3$,

$$P_c \sim \rho g R \sim \frac{M}{R^3} \frac{GM}{R^2} R = \frac{GM^2}{R^4}.$$

For Earth,

$$P_{c,\oplus} \sim \frac{GM_{\oplus}^2}{R_{\oplus}^4} \sim 10^{11}\text{--}10^{12} \text{ Pa.}$$

For the Sun,

$$P_{c,\odot} \sim \frac{GM_{\odot}^2}{R_{\odot}^4} \sim 10^{14}\text{--}10^{15} \text{ Pa.}$$

For a gas, $P \sim nk_B T \sim \rho k_B T / m_p$, so

$$k_B T_c \sim \frac{P_c}{\rho / m_p} \sim \frac{GM_{\odot} m_p}{R_{\odot}},$$

and

$$T_c \sim \frac{GM_{\odot} m_p}{k_B R_{\odot}} \sim 10^7 \text{ K},$$

which is the correct stellar-core scale.

10.8 Ideal Gas: Sound Speed and Internal Energy

Question: Given an ideal gas with pressure P and density ρ , what is the characteristic speed of disturbances and what is the internal energy density?

Solution: The only velocity that can be made from P and ρ is

$$c_s \sim \sqrt{\frac{P}{\rho}}.$$

The exact adiabatic result is $c_s = \sqrt{\gamma P / \rho}$, where γ is order unity. The internal energy density must have units of pressure, so

$$u \sim P$$

more precisely $u = P / (\gamma - 1)$ for an ideal gas.

10.9 Surface Waves

Question: Given an incompressible liquid of density ρ with a free surface, what is the speed of waves on the surface?

Solution: Density alone cannot determine a wave speed; one also needs the restoring mechanism and a length scale. For ordinary long gravity waves, the restoring acceleration is g . If the relevant horizontal wavelength is λ and the water is deep, dimensional analysis gives

$$v \sim \sqrt{g\lambda}$$

up to factors of 2π . In shallow water of depth h , the depth replaces the wavelength:

$$v \sim \sqrt{gh}.$$

If surface tension σ is the restoring force instead of gravity, the capillary-wave estimate is

$$v \sim \sqrt{\frac{\sigma}{\rho\lambda}}.$$

10.10 Relativistic Acceleration in an Electric Field

Question: A particle of charge q is accelerated by an electric field E for time t and may become relativistic. What dimensionless number characterizes the problem? What is the most general expression for the energy?

Solution: The field gives momentum

$$p \sim qEt.$$

The relativistic momentum scale is mc , so the dimensionless parameter is

$$\chi = \frac{qEt}{mc}.$$

The most general energy scale is therefore

$$\mathcal{E}(t) = mc^2 f\left(\frac{qEt}{mc}\right).$$

For a particle starting from rest in a constant field, the exact relativistic form is

$$\mathcal{E}_{\text{total}} = mc^2 \sqrt{1 + \chi^2}, \quad K = mc^2 \left(\sqrt{1 + \chi^2} - 1 \right).$$

For $\chi \ll 1$, $K \simeq (qEt)^2/(2m)$; for $\chi \gg 1$, $K \simeq qEt c$.

10.11 Dense Quantum Electrons

Question: Electrons have number density n , charge e , and mass m . The system is quantum mechanical and may be relativistic. What dimensionless numbers characterize it, and what is the general characteristic energy?

Solution: The mean interparticle spacing is

$$a \sim n^{-1/3}.$$

Quantum mechanics introduces $\hbar$, relativity introduces c , and electromagnetism introduces e^2 . A convenient pair of dimensionless numbers is

$$\alpha = \frac{e^2}{\hbar c}, \quad n\lambda_C^3 = n \left(\frac{\hbar}{mc} \right)^3,$$

where α measures electromagnetic coupling and $n\lambda_C^3$ measures whether the electron spacing is comparable to the Compton wavelength.

The most general characteristic energy can be written as

$$\mathcal{E} = mc^2 F(\alpha, n\lambda_C^3).$$

Useful limiting scales are:

$$E_F^{\text{nonrel}} \sim \frac{\hbar^2 n^{2/3}}{m}, \quad E_F^{\text{rel}} \sim \hbar c n^{1/3}, \quad E_C \sim e^2 n^{1/3}.$$

Which one dominates depends on density and coupling.

10.12 Surface Tension Versus Gravity

Question: A system contains liquid of density ρ and surface tension σ in a gravitational field, with characteristic size ℓ . What dimensionless number can be formed, what does it mean, and where is it important?

Solution: Compare gravitational energy density over height ℓ , of order $\rho g \ell$, with capillary pressure σ/ℓ . Their ratio is

$$\text{Bo} = \frac{\rho g \ell^2}{\sigma},$$

the Bond number.

- $\text{Bo} \ll 1$: surface tension dominates; drops stay round and small insects can deform the surface without sinking.
- $\text{Bo} \gg 1$: gravity dominates; large puddles flatten and waves behave mostly as gravity waves.

The crossover length $\ell_c = \sqrt{\sigma/(\rho g)}$ is the capillary length, about a few millimeters for water.

10.13 Honey Flow and Inclined-Plane Flow

Question: Define a dimensionless ratio of gravitational to viscous forces. For honey with $\nu \approx 10^{-3} \text{ m}^2/\text{s}$, what flow width is needed for it to flow freely? For steady flow on an inclined plane of angle α and thickness a , what is the most general velocity profile?

Solution: For gravity-driven flow with characteristic size L , the natural velocity from gravity is $U \sim \sqrt{gL}$. The Reynolds-like ratio of inertia/gravity-driven motion to viscous diffusion is

$$\text{Ga} \sim \frac{UL}{\nu} \sim \frac{\sqrt{gL}L}{\nu} = \frac{\sqrt{g}L^{3/2}}{\nu}.$$

Equivalently $\text{Ga}^2 = gL^3/\nu^2$. Freely flowing behavior begins when this is order unity:

$$L \sim \left(\frac{\nu^2}{g}\right)^{1/3} \sim \left(\frac{10^{-6}}{10}\right)^{1/3} \text{ m} \sim 5 \times 10^{-3} \text{ m}.$$

Thus honey needs a width/depth of order a few millimeters before gravity beats viscosity efficiently.

For steady laminar flow down an inclined plane, the dimensional form is

$$u(y) = \frac{ga^2 \sin \alpha}{\nu} f\left(\frac{y}{a}\right).$$

With no slip at the wall and zero shear at the free surface, the exact parabolic profile is

$$u(y) = \frac{g \sin \alpha}{2\nu} (2ay - y^2), \quad 0 \leq y \leq a.$$

10.14 Blood Flow: Reynolds and Womersley Numbers

Question: Pulsatile blood flow is characterized by Reynolds number and Womersley number $\alpha_W = L\sqrt{\omega/\nu}$. What do they mean, what are their orders of magnitude in the human body, and how can elasticity be included?

Solution: The Reynolds number is

$$\text{Re} = \frac{UL}{\nu},$$

the ratio of inertial advection to viscous momentum diffusion. The Womersley number is

$$\alpha_W = L\sqrt{\frac{\omega}{\nu}} = \sqrt{\frac{L^2/\nu}{1/\omega}},$$

so it compares the viscous diffusion time across the vessel to the heartbeat period.

For the aorta, take $L \sim 1$ cm, $U \sim 0.5$ m/s, $\nu \sim 3 \times 10^{-6}$ m²/s, and $\omega \sim 2\pi$ s⁻¹:

$$\text{Re} \sim \frac{0.5 \times 10^{-2}}{3 \times 10^{-6}} \sim 2 \times 10^3, \quad \alpha_W \sim 10^{-2} \sqrt{\frac{6}{3 \times 10^{-6}}} \sim 10.$$

So large arteries have important inertia and pulsatility. In capillaries, with $L \sim 5 \mu\text{m}$ and $U \sim 10^{-3}$ m/s,

$$\text{Re} \sim 10^{-3}, \quad \alpha_W \ll 1,$$

so viscosity dominates and the flow is essentially quasi-steady.

Elasticity can be represented by comparing fluid dynamic pressure to wall elastic stiffness. A useful dimensionless number is

$$\mathcal{E} = \frac{\rho U^2 R}{Eh},$$

where E is the wall Young modulus, h is wall thickness, and R is vessel radius. If $\mathcal{E} \ll 1$, the wall is effectively rigid; if it is not small, pressure pulses strongly deform the vessel.

10.15 Magnus Force on a Soccer Ball

Question: A spinning ball moves in air with speed v and angular speed ω . What dimensionless numbers characterize the Magnus force? Estimate them for a soccer kick, estimate the force, and estimate the horizontal bending angle.

Solution: The variables are $F_M, \rho, R, v, \omega, \nu$. Buckingham's theorem gives

$$\left[\text{Re} = \frac{vR}{\nu} \right], \quad \left[S = \frac{\omega R}{v} \right],$$

where S is the spin parameter. The general force is

$$\left[F_M = \rho v^2 R^2 f(\text{Re}, S) \right].$$

For high Reynolds number and moderate spin, f is roughly proportional to S , so

$$F_M \sim \rho v^2 R^2 \frac{\omega R}{v}.$$

For a soccer ball, take $R \sim 0.11$ m, $v \sim 25$ m/s, $\omega \sim 50$ rad/s, $\rho_{\text{air}} \sim 1$ kg/m³, and $\nu_{\text{air}} \sim 1.5 \times 10^{-5}$ m²/s:

$$\text{Re} \sim \frac{25 \times 0.11}{1.5 \times 10^{-5}} \sim 2 \times 10^5, \quad S \sim \frac{50 \times 0.11}{25} \sim 0.2.$$

The force scale is

$$F_M \sim 1 \times 25^2 \times (0.11)^2 \times 0.2 \sim 1\text{--}2 \text{ N}.$$

For a ball mass $m \sim 0.43$ kg, the sideways acceleration is $a_{\perp} \sim 3\text{--}5$ m/s². Over a one-second flight,

$$y \sim \frac{1}{2} a_{\perp} t^2 \sim 1\text{--}3 \text{ m}.$$

For a 20–30 m kick this is a bending angle

$$\theta \sim \frac{y}{L} \sim 0.05\text{--}0.1 \text{ rad} \sim 3^{\circ}\text{--}6^{\circ},$$

large enough to be clearly visible.

10.16 Beirut Explosion and Ambient Pressure

Question: Use videos of the Beirut explosion to estimate the released energy. If ambient air pressure matters, what dimensionless number describes the explosion? At what radius is it order unity for Beirut, Hiroshima, and the largest human-made explosion?

Solution: For a strong spherical blast wave, the Sedov-Taylor scaling from the lectures is

$$r(t) \sim \left(\frac{Et^2}{\rho} \right)^{1/5}, \quad \boxed{E \sim \rho \frac{r^5}{t^2}}.$$

From public videos of the Beirut blast, an order-of-magnitude reading is that the visible shock reached a few hundred meters in about a second. Taking $r \sim 300$ m, $t \sim 1$ s, and $\rho \sim 1$ kg/m³,

$$E \sim 1 \cdot \frac{(300)^5}{1^2} \text{ J} \sim 2 \times 10^{12} \text{ J}.$$

Since 1 kiloton TNT $\simeq 4.2 \times 10^{12}$ J, this is of order

$$E_{\text{Beirut}} \sim 0.5\text{--}2 \text{ kt TNT},$$

depending on which video frames and radii are used.

Ambient pressure becomes important when the blast energy is comparable to the work needed to push aside the surrounding atmosphere:

$$\Pi_p = \frac{p_0 r^3}{E}.$$

The strong-shock Sedov regime requires $\Pi_p \ll 1$. The crossover radius is

$$\boxed{r_p \sim \left(\frac{E}{p_0} \right)^{1/3}}, \quad p_0 \sim 10^5 \text{ Pa}.$$

For Beirut, with $E \sim 10^{12}\text{--}10^{13}$ J,

$$r_p \sim 200\text{--}500 \text{ m}.$$

For Hiroshima, $E \simeq 15 \text{ kt TNT} \simeq 6 \times 10^{13} \text{ J}$:

$$r_p \sim \left(\frac{6 \times 10^{13}}{10^5} \right)^{1/3} \sim 800 \text{ m.}$$

For the largest human-made explosion, about 57 Mt TNT,

$$E \simeq 57 \times 10^6 \times 4.2 \times 10^9 \text{ J} \simeq 2.4 \times 10^{17} \text{ J},$$

so

$$r_p \sim \left(\frac{2.4 \times 10^{17}}{10^5} \right)^{1/3} \sim 10\text{--}15 \text{ km.}$$

The meaning is physical: below r_p the explosion behaves like a strong blast wave controlled mainly by E, ρ, t ; near and beyond r_p , ambient pressure is no longer negligible, and the shock weakens toward an ordinary sound wave.